L Pro

System Architecture, Multiphysics Modeling, and Closed-Loop Control of an Automated Robotic Hair-Washing Platform

L Pro Engineering Technical Paper

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Scope statement. This document is a theory-and-design paper constructed from the supplied L Pro render set. The renders clearly expose the outer enclosure, head interface, clustered wash/massage heads, dual process reservoirs, cartridge bank, pumps/tubing, drainage basin, access panels, display, and positioning arms. Exact CAD dimensions, pump curves, sensor part numbers, valve coefficients, and measured prototype data were not supplied. Therefore, the paper deliberately separates *physics* from *evidence*: the equations are first-principles engineering models, while the numerical figures are stochastic digital-twin studies used to test sensitivity, uncertainty, saturation, repeatability, and constraint violation. They include manufacturing spread, sensor noise, actuator lag, alignment error, hair-load variation, and hydraulic/thermal disturbances rather than idealized deterministic curves. None of those numerical traces is claimed as L Pro prototype measurement. Functional labels assigned to rendered components should be reconciled with the production BOM before external publication.

Abstract

L Pro is an automated hair-washing platform that integrates a conformal head interface, distributed fluid jets, compliant massage end-effectors, chemical dosing, thermal conditioning, recirculation/drainage, sensing, and closed-loop control in a compact appliance-scale enclosure. The engineering difficulty is not merely to spray water onto hair, but to guarantee spatially uniform wetting, bounded scalp contact pressure, adequate rinsing, temperature safety, drainage margin, and comfortable treatment across substantial variation in head geometry and hair state. This paper develops a first-principles model of L Pro as a coupled hydraulic-thermal-contact-robotic system. A pressure-flow network model maps pump head and valve commands to nozzle discharge; free-jet and impingement models map discharge to local momentum flux; a probabilistic surface-exposure model quantifies wash coverage; Hertz-type compliant contact equations bound massage force; mixing and residence-time equations govern shampoo and rinse concentration; and a lumped thermal model governs delivered-water temperature. These components are unified by a constrained controller and a model-predictive cycle planner that minimize water, energy, and cycle time while satisfying pressure, temperature, coverage, and fault-tolerance constraints. Several analytical properties are established, including monotonicity of cumulative coverage, a sufficient drainage-capacity inequality, bounded-temperature invariance, and closed-loop stability for the nominal flow-control subsystem. The resulting framework provides a rigorous explanation of how L Pro can transform a mechanically dense set of reservoirs, pumps, manifolds, clustered heads, and user-positioning hardware into a repeatable automated wash cycle.

Keywords: automated hair washing; robotic personal care; fluidic robotics; compliant contact; spray coverage; model predictive control; safety-constrained automation; human-machine interface.

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1 Introduction

Automated hair washing is a deceptively hard robotics problem. A human stylist continuously adapts pressure, flow, temperature, detergent amount, finger trajectory, and rinse duration using tactile and visual cues. An automatic machine must reproduce the useful part of this adaptation without direct human dexterity, and must do so under strong comfort and safety constraints. The task spans multiple physical domains: liquid transport, free jets, surface wetting, deformable contact, thermal transients, chemical dilution, geometric coverage, drainage, sensor fusion, and feedback control.

The supplied L Pro render set suggests an architecture specifically designed to make these interactions controllable. The head is inserted into a padded opening surrounded by a crown of clustered treatment heads. The chamber walls contain additional distributed modules and red illuminated structural rails. Below the treatment region, a perforated collection bowl drains liquid toward a central return. Two large process reservoirs occupy the lower body, while a smaller multi-cartridge bank supports metered additives. Internally, transparent hoses connect pumps, manifolds, filters, reservoirs, and the treatment chamber. Side-mounted articulated fixtures provide head or accessory positioning, and the rear panel incorporates a display for user interaction. These visual elements motivate the system decomposition in Fig. 1.



(a) Closed front view. (b) Open front view. (c) Side enclosure. (d) Side internal view.

Figure 1: Representative L Pro views from the supplied render set. The machine combines a conformal head interface, distributed treatment heads, internal fluid routing, removable reservoirs, and serviceable enclosure panels.

This paper has four goals. First, it maps the visible hardware into a coherent functional architecture. Second, it derives the governing equations needed to size and control the fluid, thermal, contact, and positioning subsystems. Third, it establishes analytical guarantees that are useful before prototype testing. Fourth, it defines a validation program that can turn the theoretical model into a calibrated digital twin of the production machine.

The main modeling principle is to treat washing quality as a *distributed constrained exposure problem*. At each scalp location $\boldsymbol{x}$, L Pro must deliver enough water, detergent, and mechanical action over time, but

must never violate local pressure or thermal bounds. A compact representation is

$$\mathcal{Q}(\boldsymbol{x}) = f\left(\underbrace{\int_0^T q(\boldsymbol{x},t)\,\mathrm{d}t}_{\text{hydraulic exposure}}, \underbrace{\int_0^T c_s(\boldsymbol{x},t) q(\boldsymbol{x},t)\,\mathrm{d}t}_{\text{chemical dose}}, \underbrace{\int_0^T \phi(F_n, v_t)\,\mathrm{d}t}_{\text{mechanical action}}, \underbrace{T_w(\boldsymbol{x},t)}_{\text{thermal state}} \right), \tag{1}$$

subject to comfort and safety limits. The controller therefore optimizes a field over the scalp, not a single scalar flow rate.

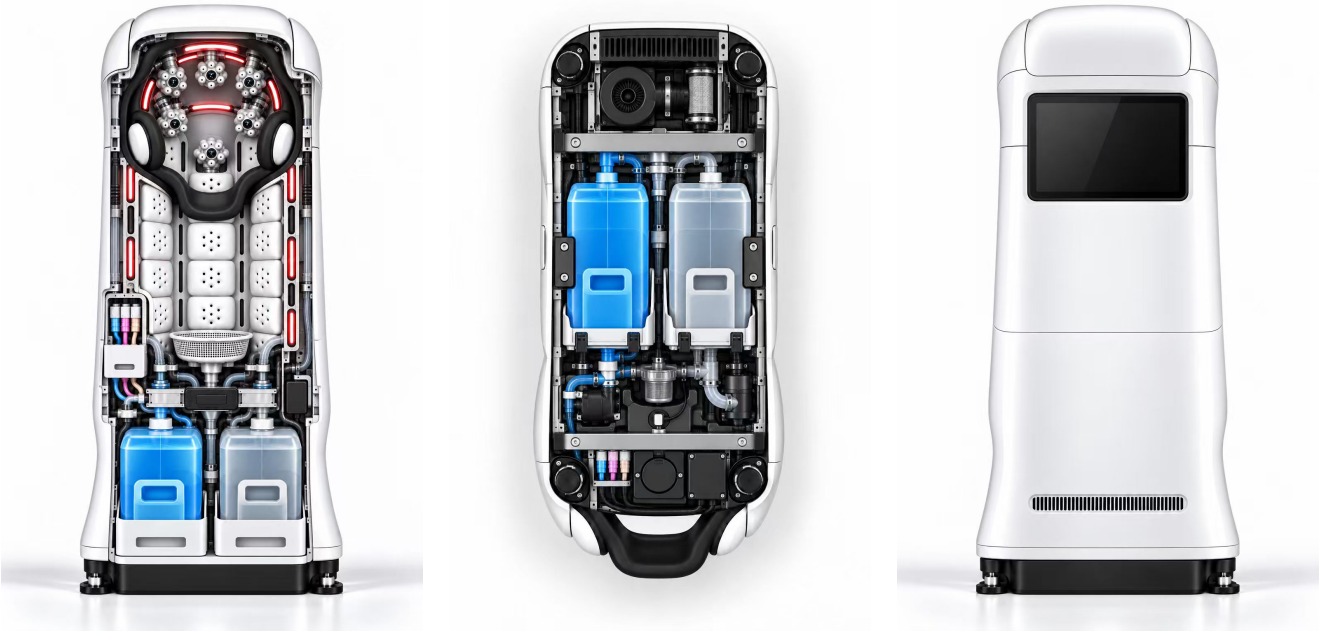
2 System Architecture and Functional Decomposition

2.1 Physical subsystems visible in the render set

A practical L Pro implementation can be partitioned into eight functional subsystems. The visible hardware supports this partition even though the exact BOM is not available.

Table 1: Functional decomposition of L Pro.

Subsystem	**Visible or implied hardware**	**Primary controlled variables**
Head interface	Padded cranial opening, chamber liner, distributed clustered heads	Head pose, clearance, contact load
Wash delivery	Pumps, hoses, manifolds, nozzles/end-effectors	Flow Q, pressure Δp, jet duty cycle
Chemical dosing	Small cartridge bank, metering lines	Shampoo/conditioner concentration c_s
Thermal conditioning	Reservoirs, heat exchanger/heater volume, temperature sensors	Delivered temperature T_w
Massage/contact	Multi-lobed compliant treatment heads	Normal force F_n, penetration δ, tangential speed v_t
Drain/recovery	Perforated basin, central drain, return hose, filter	Drain flow Q_d, sump level h_s
Positioning	Side articulated modules, head support geometry	Relative head pose $\boldsymbol{p}, \boldsymbol{R}$
Supervisory control	Rear display, embedded electronics, sensors	State sequence, alarms, user parameters



(a) Front internal packaging. (b) Rear/top internal packaging. (c) Rear user interface.

Figure 2: L Pro packaging emphasizes short fluid paths around the head chamber while placing larger reservoirs and pumps in the lower body.

Functional anatomy inferred from the supplied L Pro internal render

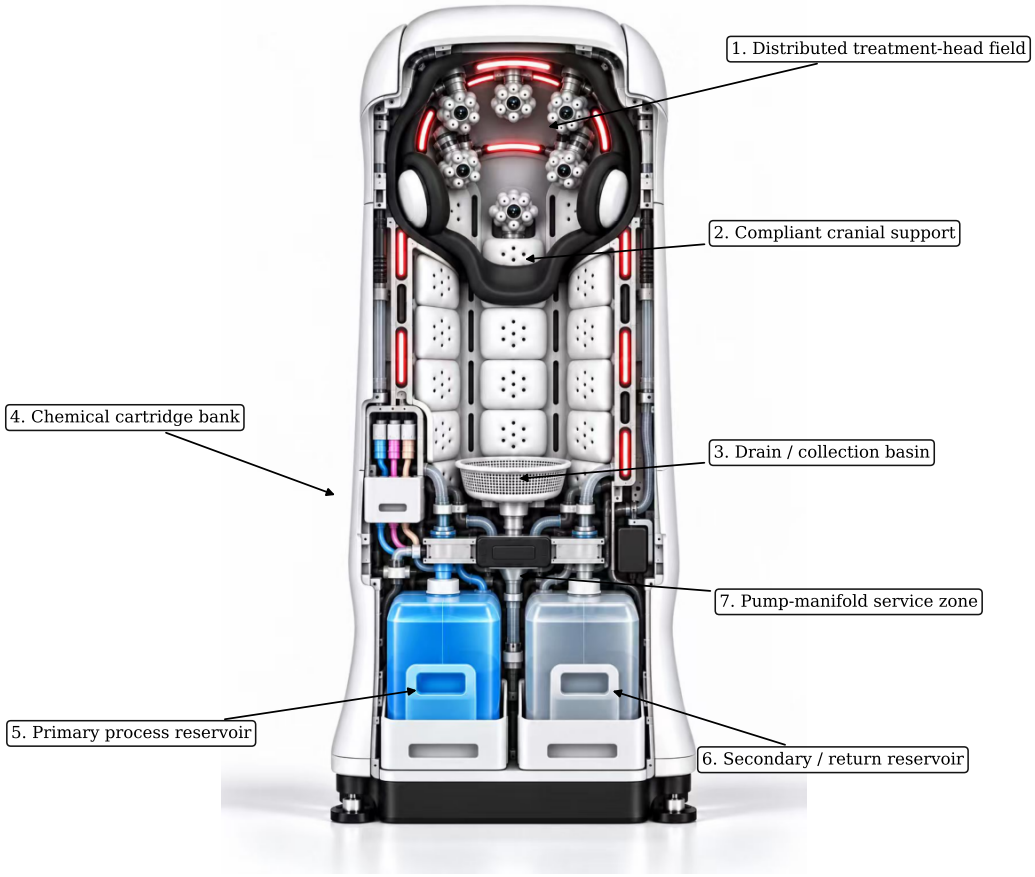


Figure 3: Annotated functional anatomy of the L Pro internal package. Numbered callouts map the supplied render to the treatment-head field, cranial support, drain basin, cartridge bank, two process reservoirs, and pump-manifold service zone. The labels are functional interpretations of the render and should be reconciled with the production BOM before external release.

2.2 Signal and energy flow

The machine can be understood as three coupled networks: a hydraulic network carrying mass and pressure, an electrical/thermal network carrying energy, and an information network carrying state estimates and control commands. The supervisory controller should not directly command "wash harder." It should command physically interpretable references such as manifold pressure, jet duty ratio, additive concentration, contact-force envelope, and target temperature.

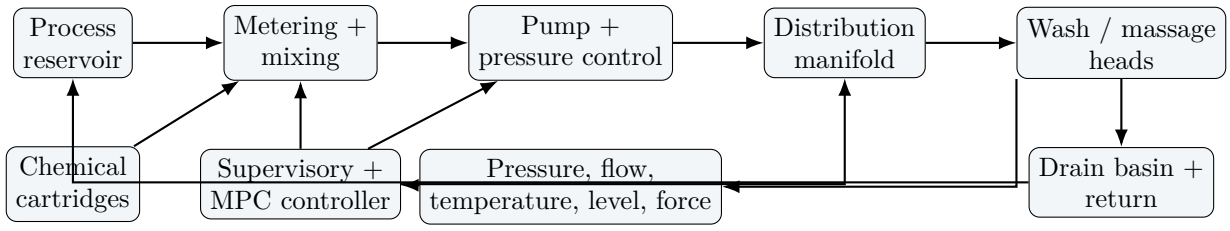


Figure 4: Functional fluid-and-control loop. The exact production plumbing may differ; this diagram expresses the control structure implied by the rendered reservoirs, pumps, manifolds, treatment heads, and drain.

3 Hydraulic Network Model

3.1 Pump-to-nozzle pressure balance

For incompressible liquid, the pump operating point is determined by the intersection of the pump curve and the system curve. Let Q denote total manifold flow. A convenient reduced system model is

$$\Delta p_p(Q, u_p) = \Delta p_{static} + K_h Q^2, \tag{2}$$

where $u_p \in [0, 1]$ is normalized pump command, Δp_p is pump head, and K_h aggregates tube, fitting, filter, valve, and manifold losses. The quadratic approximation follows the usual turbulent minor-loss relation

$$\Delta p_j = K_j \frac{\rho v_j^2}{2} = K_j \frac{\rho}{2} \left(\frac{Q}{A_j} \right)^2. \tag{3}$$

For a pipe of length L and diameter D, Darcy-Weisbach adds

$$\Delta p_f = f_D \frac{L}{D} \frac{\rho v^2}{2}, \tag{4}$$

with friction factor f_D. The Reynolds number is

$$\mathrm{Re} = \frac{\rho v D}{\mu}, \tag{5}$$

which determines whether viscous or turbulent correlations should be used during calibration.

If N treatment heads are independently valved, the total flow is

$$Q = \sum_{i=1}^{N} s_i Q_i, \qquad s_i \in [0, 1], \tag{6}$$

and the local branch flow can be approximated by the orifice equation

$$Q_i = C_{d,i} A_i \sqrt{\frac{2(p_m - p_a)}{\rho}}, \tag{7}$$

where p_m is manifold pressure and p_a is chamber pressure. This square-root relation is central to both calibration and control.

3.2 Sensitivity to valve and nozzle variation

Manufacturing variation in nozzle area and discharge coefficient changes delivered flow. Linearizing Eq. (7) around $(A_0, \Delta p_0)$ gives

$$\frac{\delta Q}{Q_0} \approx \frac{\delta C_d}{C_{d,0}} + \frac{\delta A}{A_0} + \frac{1}{2} \frac{\delta \Delta p}{\Delta p_0}. \tag{8}$$

Thus a 2% geometric area error has approximately twice the first-order flow effect of a 2% pressure error. This is a strong argument for per-head calibration coefficients in the controller.

Hydraulic digital-twin study with load variation, manufacturing tolerance, and sensing noise

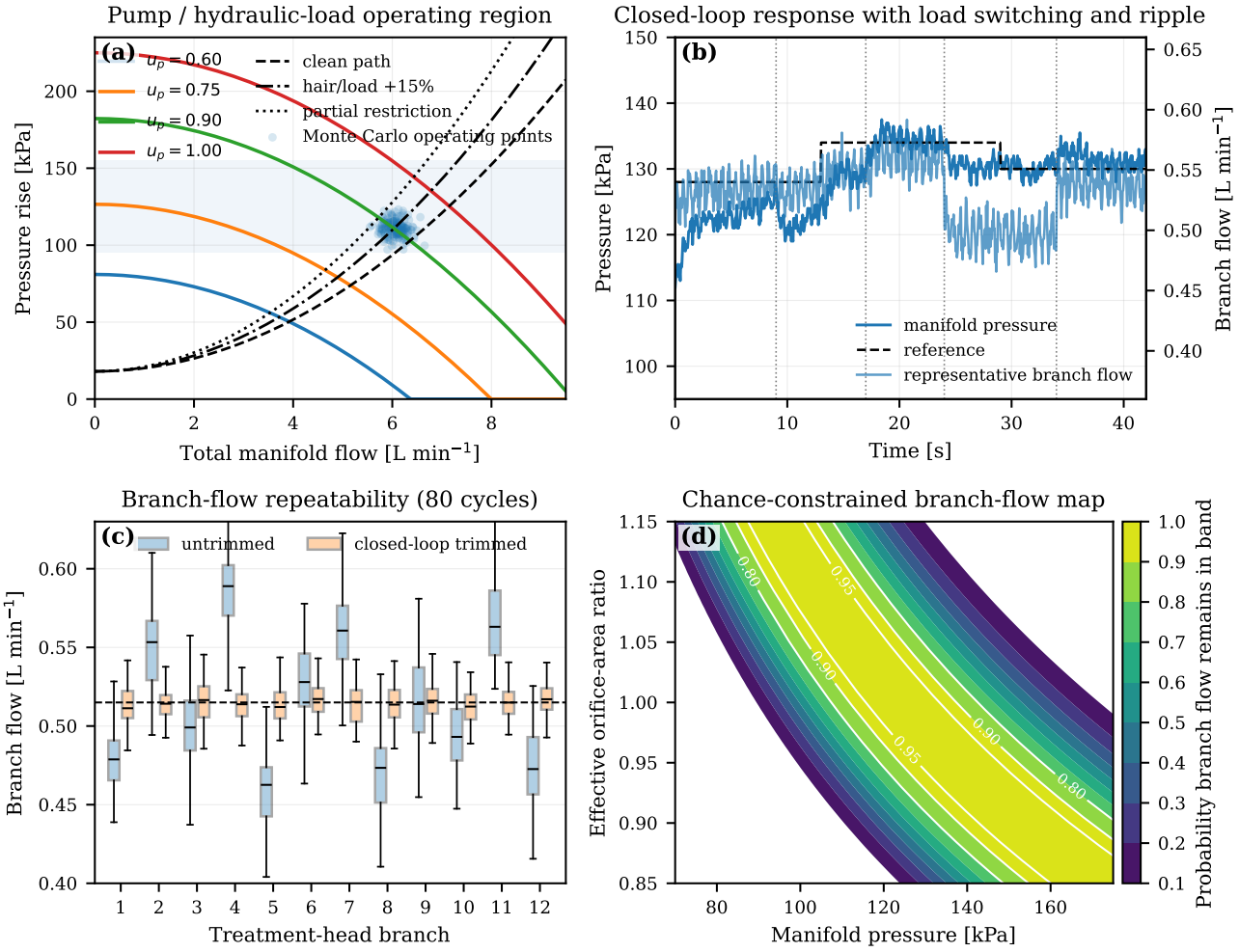


Figure 5: Stochastic hydraulic digital-twin study. (a) Pump and hydraulic-load families are shown together with a Monte Carlo operating-point cloud caused by pump-curve and load uncertainty. (b) Closed-loop manifold pressure and representative branch flow include switching disturbances, pump ripple, sensor noise, quantization, and actuator lag. (c) Boxplots summarize 80 repeated simulated cycles for twelve branches before and after valve trim. (d) A chance-constrained map reports the probability that a branch remains inside its target flow band as pressure and effective orifice area vary. The dimensional values are engineering design-study values, not measured L Pro specifications.

3.3 Manifold balancing

Let G_i denote the hydraulic conductance of branch i, where $Q_i \approx G_i\sqrt{p_m - p_a}$. Uniform flow requires G_i to be equalized. If branch resistance varies, active valve trim α_i can be chosen by solving

$$\min_{\alpha_1,\dots,\alpha_N} \sum_{i=1}^{N} (Q_i - Q^\star)^2 + \lambda \sum_{i=1}^{N} (\alpha_i - \alpha_{i,0})^2, \tag{9}$$

subject to $0 \le \alpha_i \le 1$. This turns manifold balancing into a simple convex calibration problem after local linearization.

4 Jet Impingement, Wetting, and Scalp-Surface Loading

4.1 Free-jet velocity and momentum flux

At a nozzle with exit area A_n, the exit velocity is

$$v_0 = \frac{Q_n}{A_n}. \tag{10}$$

A free jet entrains surrounding air and broadens with distance. A compact engineering approximation for centerline velocity is

$$v_c(z) = \frac{v_0}{1 + k_e z/D_n}, \tag{11}$$

where k_e is an entrainment constant fitted from bench data. The maximum normal momentum pressure on a surface at incidence angle θ is modeled as

$$p_{imp}(z, \theta) = C_i \frac{1}{2} \rho v_c^2(z) \cos^2 \theta. \tag{12}$$

The controller can therefore reduce scalp loading by lowering pressure, increasing standoff, or changing incidence.

The liquid momentum flux is

$$\dot{\Pi} = \rho Q v, \tag{13}$$

and the reaction force on a local surface patch is approximately

$$F_j \approx \rho Q v (1 - \cos \psi), \tag{14}$$

where ψ describes deflection of the outgoing film.

4.2 Distributed wetting kernel

A single nozzle does not wet a point only at its centerline. Define a normalized spatial kernel

$$g_i(\boldsymbol{x}, t) = \exp\left[-\frac{\|\boldsymbol{x} - \boldsymbol{x}_i(t)\|^2}{2\sigma_i^2} \right] \chi_i(\boldsymbol{x}, t), \tag{15}$$

where σ_i captures spread and χ_i is a visibility/occlusion factor accounting for hair and geometry. Local hydraulic exposure is

$$H(\boldsymbol{x}, T) = \int_0^T \sum_{i=1}^{N} \lambda_i(t) g_i(\boldsymbol{x}, t) \, \mathrm{d}t, \tag{16}$$

with $\lambda_i \propto Q_i$ or $\lambda_i \propto \rho Q_i v_i$, depending on whether wetting or momentum is the targeted mechanism.

A saturation model maps exposure to coverage probability:

$$C(\boldsymbol{x}, T) = 1 - e^{-H(\boldsymbol{x}, T)}. \tag{17}$$

This simple nonlinear transformation is useful because it captures diminishing returns: repeatedly washing the same area gives less benefit than redirecting treatment toward an underexposed region.

Proposition 1 (Monotonic cumulative coverage)**.** *If $\lambda_i(t) \geq 0$ and $g_i(\boldsymbol{x}, t) \geq 0$, then $C(\boldsymbol{x}, T)$ in Eq. (17)*

is nondecreasing in T and bounded in $[0, 1)$.

Proof. Since $\dot{H}(\boldsymbol{x}, T) = \sum_i \lambda_i(T) g_i(\boldsymbol{x}, T) \geq 0$, we have

$$\frac{\partial C}{\partial T} = e^{-H} \dot{H} \geq 0.$$

Also $H \geq 0$, so $0 \leq 1 - e^{-H} < 1$. □

4.3 Hair-layer hydraulic resistance

Hair modifies flow by adding a porous, deformable layer between jet and scalp. A Darcy-type model is

$$\boldsymbol{u}_h = -\frac{k_h}{\mu} \nabla p, \tag{18}$$

where effective permeability k_h depends on hair density, wetting, compression, and local orientation. For a layer thickness L_h, the pressure drop becomes

$$\Delta p_h \approx \frac{\mu L_h}{k_h} u_h. \tag{19}$$

This motivates adaptive pressure: thick or dense hair can require greater hydraulic head, but only within the scalp-pressure envelope.

5 Head Geometry and Coverage Planning

5.1 Geometric scalp model

For control, a person's head can be approximated by an ellipsoid with semi-axes a, b, c:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1. \tag{20}$$

A more accurate implementation would fit a low-resolution point cloud or depth map to a deformation field $\boldsymbol{d}(u, v)$ over a template mesh. The surface normal at a mesh vertex $\boldsymbol{x}_k$ is $\boldsymbol{n}_k$, and the nozzle-to-surface incidence is

$$\cos \theta_{ik} = \frac{(\boldsymbol{x}_k - \boldsymbol{o}_i)^T \boldsymbol{n}_k}{\|\boldsymbol{x}_k - \boldsymbol{o}_i\|}. \tag{21}$$

Nozzle i is considered geometrically effective at point k if

$$d_{min} \leq \|\boldsymbol{x}_k - \boldsymbol{o}_i\| \leq d_{max}, \qquad \theta_{ik} \leq \theta_{max}. \tag{22}$$

5.2 Protected and comfort zones

The scalp mask $\mathcal{S}$ excludes protected regions such as eyes, face, and ears. A signed-distance safety margin can be written

$$d_{safe}(\boldsymbol{x}, \mathcal{P}) \geq d_0, \tag{23}$$

where $\mathcal{P}$ is the union of protected zones. Any mobile head or high-pressure jet must satisfy this constraint over its planned trajectory.

5.3 Coverage optimization

Discretize the scalp into M surface cells and the treatment sequence into K time intervals. Let $a_{mik} \geq 0$ be exposure delivered to cell m by head i during interval k at unit actuation. The planner chooses $u_{ik} \in [0, 1]$ and can solve

$$\min_{\boldsymbol{u}} \quad \alpha T + \beta V_w + \gamma E + \lambda \sum_{m=1}^{M} \xi_m^2, \tag{24}$$

$$\text{s.t.} \quad \sum_{i,k} a_{mik} u_{ik} + \xi_m \geq H_m^{min}, \quad \xi_m \geq 0, \tag{25}$$

$$p_{imp,mk} \leq p_m^{max}, \quad T_{min} \leq T_{mk} \leq T_{max}, \tag{26}$$

$$0 \leq u_{ik} \leq 1. \tag{27}$$

This structure is particularly suitable for model predictive control (MPC): after each sensor update, the remaining cycle is replanned using the current exposure map.

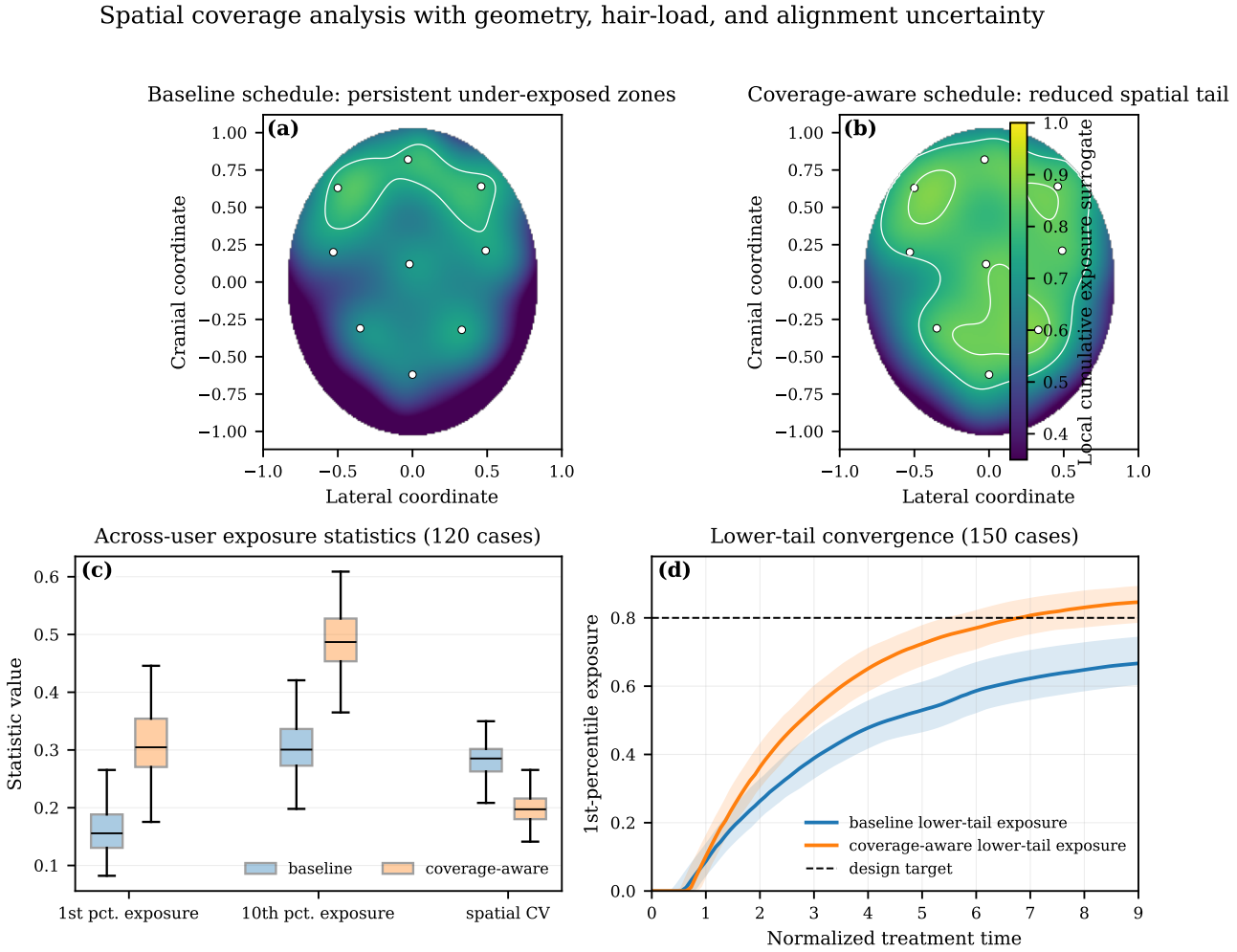


Figure 6: Robust spatial-coverage analysis generated from Eqs. (15) to (17). Panels (a)–(b) include asymmetric hair resistance, local shielding, anisotropic spray footprints, and small-scale spatial roughness rather than an ideal radial field. Panel (c) reports the distribution of lower-tail exposure and spatial coefficient of variation across 120 randomized head-geometry, hair-load, and alignment realizations. Panel (d) shows the median and 10–90% envelope of lower-tail coverage over 150 simulated cycles. These are stochastic design-study outputs, not user measurements.

6 Compliant Massage and Contact Mechanics

6.1 Why compliance is essential

The clustered white treatment heads visible around the cranial chamber are naturally modeled as compliant, multi-point scalp interfaces. Compliance serves two purposes: it distributes load over uncertain head geometry and it enables tactile massage without requiring millimeter-perfect geometric registration.

For a rounded elastic contact of effective radius R indenting the scalp by δ, Hertzian contact provides the approximate force law

$$F_n = \frac{4}{3} E^* \sqrt{R}\, \delta^{3/2}, \tag{28}$$

where the effective modulus is

$$\frac{1}{E^*} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}. \tag{29}$$

The contact radius is

$$a_c = \left(\frac{3F_n R}{4E^*}\right)^{1/3}, \tag{30}$$

and peak pressure is

$$p_0 = \frac{3F_n}{2\pi a_c^2}. \tag{31}$$

Although biological tissue is viscoelastic and hair changes contact geometry, these equations remain useful for conservative force envelope design.

6.2 Viscoelastic correction

A Kelvin-Voigt approximation adds rate dependence:

$$F_n(t) = k_c\delta(t) + b_c\dot{\delta}(t), \tag{32}$$

which can be identified experimentally from slow and fast indentation tests. To prevent an abrupt force spike during initial contact, a reference can be shaped as

$$F_n^\star(t) = F_{max}\left(3s^2 - 2s^3\right), \qquad s = \mathrm{clip}\left(\frac{t}{t_r}, 0, 1\right). \tag{33}$$

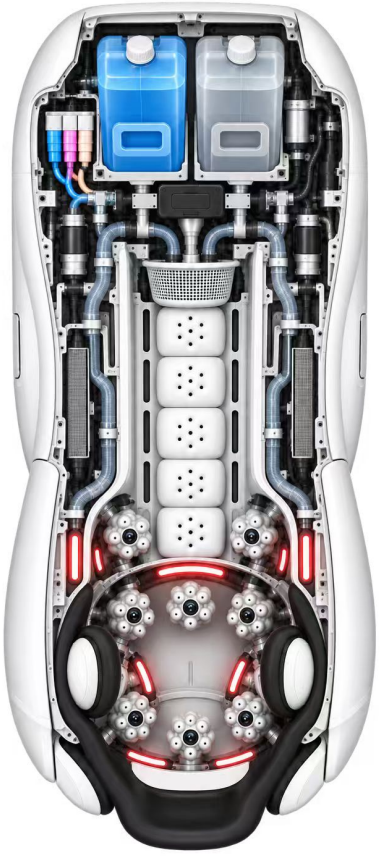
6.3 Tangential massage and slip

If a treatment head executes an orbital or rotary motion, tangential traction should remain below a comfort-limited friction envelope

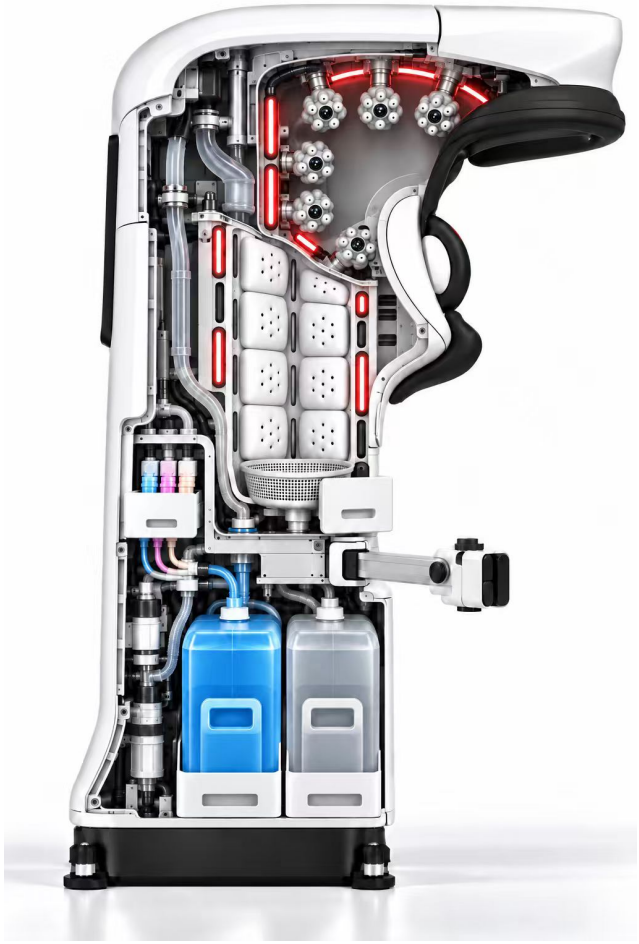
$$F_t \le \mu_c F_n, \tag{34}$$

where μ_c is an effective wet hair/scalp friction coefficient. A smooth traction objective is

$$J_{massage} = \int_0^T \left[w_F(F_n - F_n^\star)^2 + w_v(v_t - v_t^\star)^2 + w_s \max(0, F_t - \mu_c F_n)^2\right] \mathrm{d}t. \tag{35}$$



(a) Top/interior view of the distributed head field.



(b) Side/interior view showing chamber depth and lower fluid hardware.

Figure 7: The multi-head layout provides spatial redundancy: several heads can contribute to a region while individual heads remain within pressure or motion limits.

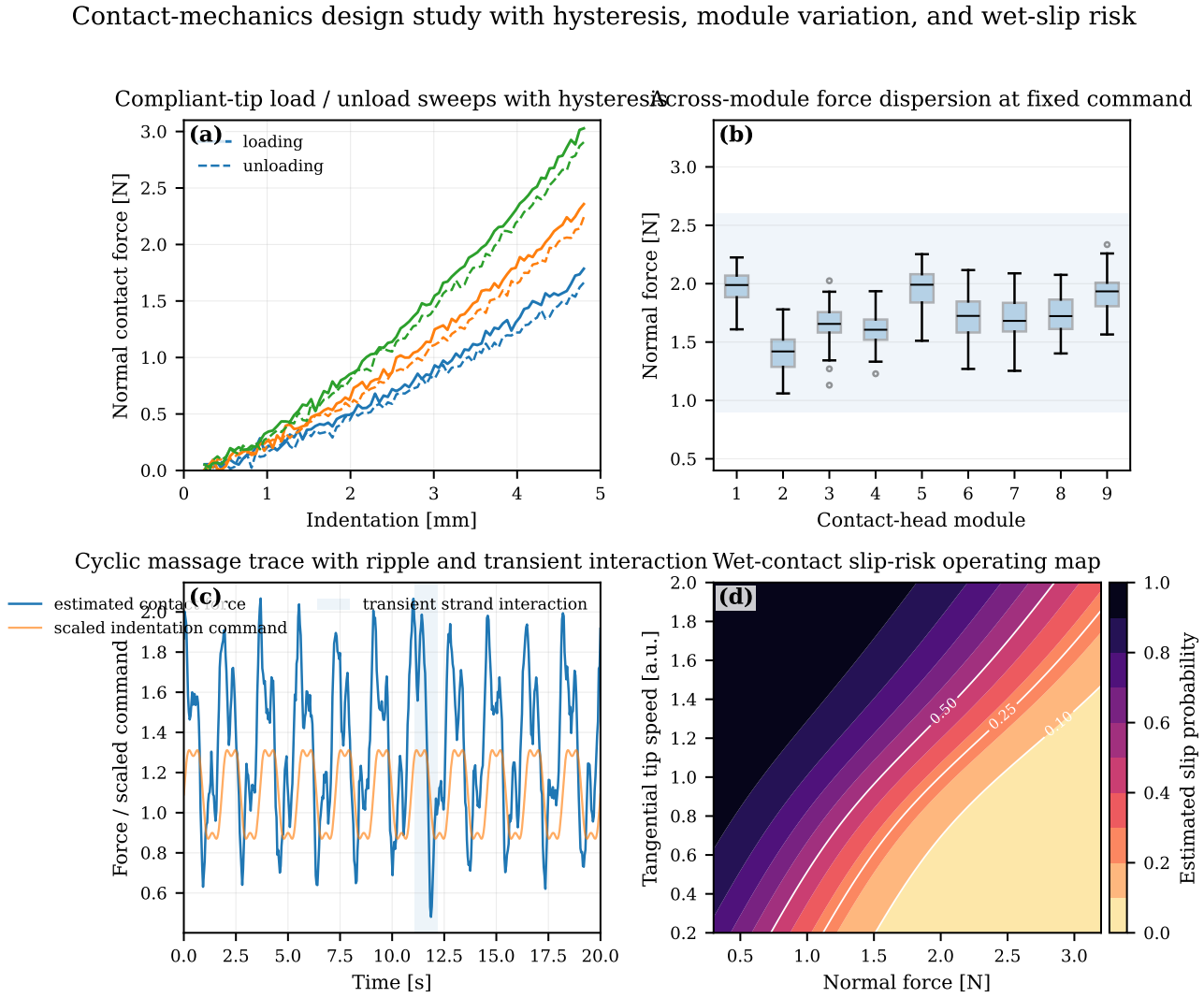


Figure 8: Non-ideal compliant-contact design study. (a) Noisy loading/unloading sweeps expose stiffness dependence and hysteresis rather than a single analytic Hertz curve. (b) Module-level boxplots show dispersion in normal force under a common command. (c) A cyclic massage trace includes drive ripple and a short strand-interaction transient. (d) A wet-contact map converts normal load and tangential tip speed into estimated slip probability. The study is intended for calibration and safety-margin design, not as reported human-subject data.

7 Chemical Dosing and Mixing

7.1 Mass balance

Let water flow be Q_w, additive flow be Q_s, and concentrate fraction be $c_{s,0}$. After ideal mixing, delivered concentration is

$$c_s = \frac{Q_s c_{s,0}}{Q_w + Q_s}. \tag{36}$$

If the desired concentration is $c_s^\star$, the required additive-to-water ratio is

$$\frac{Q_s}{Q_w} = \frac{c_s^\star}{c_{s,0} - c_s^\star}. \tag{37}$$

This ratio control is more robust than commanding a fixed dosing-pump speed because main water flow may vary across wash phases.

7.2 Mixer dynamics

A finite mixing volume V_m obeys

$$V_m \dot{c}_m = Q_{in}(c_{in} - c_m), \tag{38}$$

assuming constant volume and perfect mixing. The concentration time constant is

$$\tau_m = \frac{V_m}{Q_{in}}. \tag{39}$$

Therefore, phase transitions should account for transport delay. If rinse begins immediately after closing the dosing valve, residual additive persists for several τ_m. A useful rinse criterion is

$$c_m(t) \le c_{rinse}^{max} \quad \Rightarrow \quad t \ge \tau_m \ln \frac{c_m(0)}{c_{rinse}^{max}}. \tag{40}$$

7.3 Multi-cartridge selection

The rendered cartridge bank can support multiple formulas. If r_j denotes the fraction drawn from cartridge j, then

$$\boldsymbol{c}_{out} = \frac{Q_w \boldsymbol{c}_w + \sum_{j=1}^{J} Q_j \boldsymbol{c}_j}{Q_w + \sum_j Q_j}, \tag{41}$$

where $\boldsymbol{c}$ can represent surfactant, conditioner, fragrance, or other allowed components. Operationally, this becomes a constrained blending problem with cartridge-level bounds and contamination-avoidance purge steps.

Chemical dosing and sanitation under transport delay, pulsation, and dead-volume uncertainty

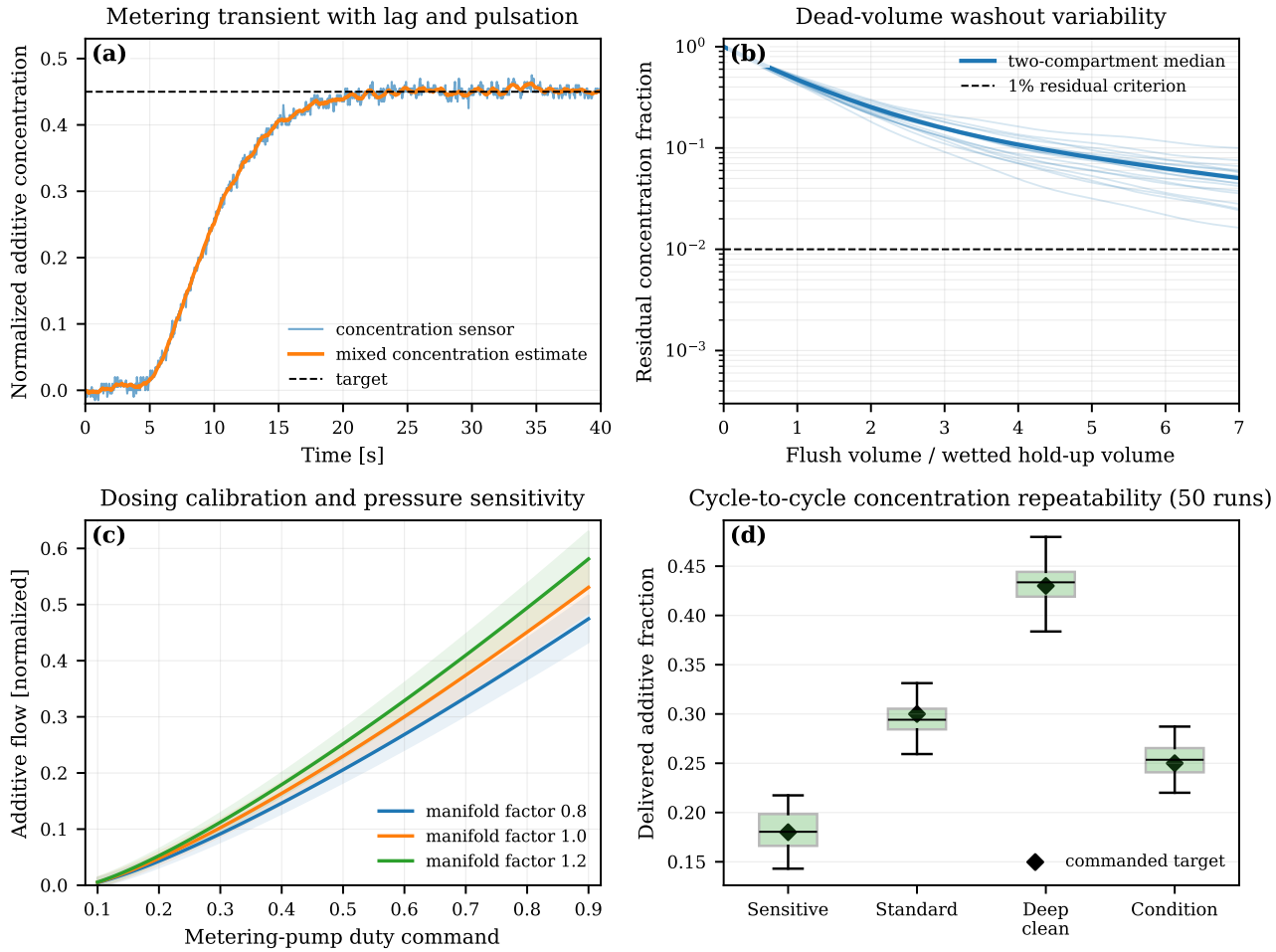


Figure 9: Stochastic dosing and sanitation study. (a) Metering dynamics include transport delay, finite-volume mixing, periodic pump pulsation, sensor noise, and quantization. (b) Eighteen dead-volume realizations show how slow compartments broaden the sanitation washout tail. (c) Nonlinear dosing calibration is shown with pressure-dependent uncertainty bands. (d) Fifty repeated cycles per recipe expose concentration repeatability around the commanded target. Values are computational design-study values rather than measurements from the production fluid path.

8 Thermal Conditioning

8.1 Lumped thermal model

Comfort depends strongly on delivered-water temperature. Let a well-mixed conditioning volume have temperature T, liquid mass m, inlet mass flow $\dot{m}$, inlet temperature T_{in}, heater power P_h, and overall heat loss coefficient UA. The energy balance is

$$mc_p\dot{T} = \dot{m}c_p(T_{in} - T) + P_h - UA(T - T_a). \tag{42}$$

For constant inputs, the steady state is

$$T_{ss} = \frac{\dot{m}c_p T_{in} + P_h + UAT_a}{\dot{m}c_p + UA}, \tag{43}$$

and the thermal time constant is

$$\tau_T = \frac{mc_p}{\dot{m}c_p + UA}. \tag{44}$$

Proposition 2 (Temperature invariant set)**.** *Suppose the controller enforces $P_h \in [0, P_{max}]$, and for all $T = T_{max}$ the right-hand side of Eq. (42) is non-positive, while for all $T = T_{min}$ it is non-negative. Then the interval $[T_{min}, T_{max}]$ is forward invariant.*

Proof. At the upper boundary the vector field points inward or tangent because $\dot{T} \leq 0$; at the lower boundary it points inward or tangent because $\dot{T} \geq 0$. A continuous first-order ODE cannot leave the interval without crossing a boundary with outward-pointing derivative, which is excluded. □

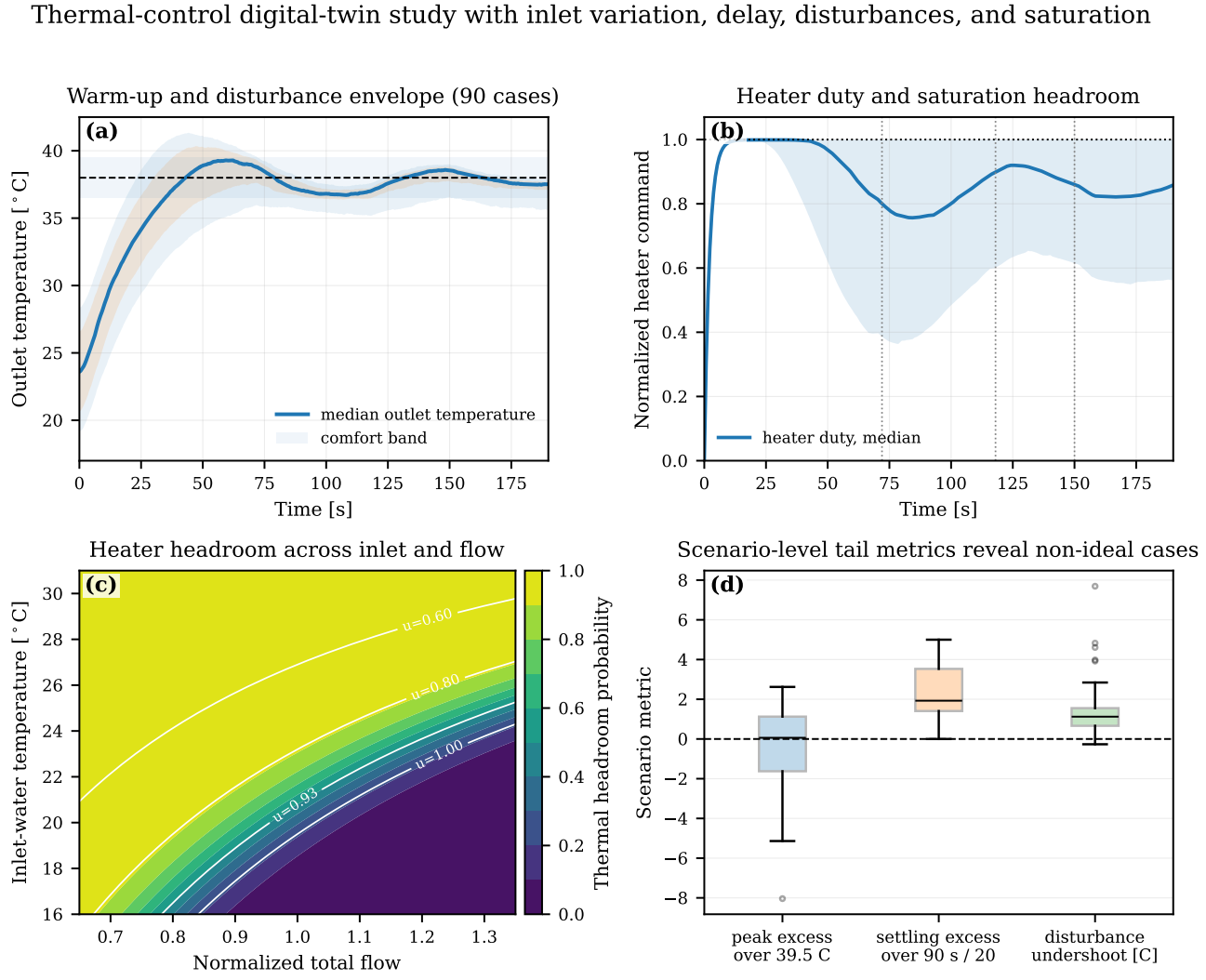


Figure 10: Thermal-control uncertainty study over 90 randomized scenarios. (a) Median outlet temperature is surrounded by nested scenario envelopes generated by inlet-temperature spread, transport delay, thermal time-constant variation, sensing noise, and flow disturbances. (b) Heater-duty trajectories reveal actuator saturation and reduced headroom during disturbances. (c) The inlet-temperature/flow map reports thermal headroom rather than an ideal steady-state temperature surface. (d) Scenario-level tail metrics retain overshoot, slow settling, and disturbance undershoot instead of suppressing adverse cases.

9 Drainage, Recovery, and Overflow Prevention

9.1 Basin mass balance

The visible perforated bowl beneath the treatment chamber acts as a distributed collector. Let basin liquid volume be V_b. Then

$$\dot{V}_b = Q_{in} - Q_d - Q_{carry}, \tag{45}$$

where Q_{in} is aggregate wash inflow, Q_d is drain flow, and Q_{carry} is liquid carried away by hair or surfaces.

Proposition 3 (Sufficient no-accumulation drainage condition)**.** *If for every admissible treatment*

command

$$Q_d^{min} \geq (1+\eta) Q_{in}^{max} \tag{46}$$

for some $\eta > 0$, then while the drain remains unobstructed the basin volume cannot increase under worst-case inflow, neglecting initial transients and storage in hoses.

Proof. From Eq. (45), $\dot{V}_b \leq Q_{in}^{max} - Q_d^{min} \leq -\eta Q_{in}^{max} < 0$, after neglecting the nonnegative Q_{carry}. □

In practice the drain controller should use a level sensor and a two-threshold policy: above warning level h_1, treatment flow is reduced; above critical level h_2, all inlet valves close and the drain pump runs at maximum safe command.

9.2 Filter pressure-drop monitoring

A clogged filter increases hydraulic resistance. If p_u and p_d are upstream and downstream pressures, a normalized clog indicator is

$$\zeta_f = \frac{(p_u - p_d)/Q^2}{K_{f,clean}}, \tag{47}$$

with maintenance requested when ζ_f exceeds a calibrated threshold.

10 Sensing and State Estimation

10.1 State vector

A minimal estimator state can include

$$\boldsymbol{x} = \begin{bmatrix} p_m & Q & T & c_m & h_b & \boldsymbol{F}_n^T & \boldsymbol{z}_h^T & \boldsymbol{b}^T \end{bmatrix}^T, \tag{48}$$

where $\boldsymbol{z}_h$ parameterizes head pose/geometry and $\boldsymbol{b}$ contains slowly varying sensor biases. Sensor outputs are

$$\boldsymbol{y}_k = h(\boldsymbol{x}_k, \boldsymbol{u}_k) + \boldsymbol{v}_k. \tag{49}$$

An extended Kalman filter (EKF) uses

$$\hat{\boldsymbol{x}}_{k|k-1} = f(\hat{\boldsymbol{x}}_{k-1|k-1}, \boldsymbol{u}_{k-1}), \tag{50}$$

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^T + Q_k, \tag{51}$$

$$K_k = P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1}, \tag{52}$$

$$\hat{\boldsymbol{x}}_{k|k} = \hat{\boldsymbol{x}}_{k|k-1} + K_k(\boldsymbol{y}_k - h(\hat{\boldsymbol{x}}_{k|k-1}, \boldsymbol{u}_k)). \tag{53}$$

The estimator is especially useful because several quantities are difficult to sense directly. For example, delivered concentration can be estimated from dosing-pump flow and mixer dynamics even if no in-line concentration sensor is present.

10.2 Redundant plausibility checks

The machine should compare redundant physical balances. Examples include:

$$r_Q = Q_{pump} - \sum_i Q_i, \tag{54}$$

$$r_V = V_{tank}(k) - V_{tank}(k-1) + \int Q_{out}\mathrm{d}t, \tag{55}$$

$$r_T = T_{out} - \hat{T}_{out}, \tag{56}$$

$$r_F = F_{meas} - \hat{F}(\delta, \dot{\delta}). \tag{57}$$

Persistent large residuals indicate leakage, blockage, dry running, sensor failure, or unexpected user motion.

11 Closed-Loop Pressure and Flow Control

11.1 Local plant model

Around an operating point, pump/manifold dynamics can often be reduced to

$$\tau_p \dot{p}_m + p_m = K_p u_p - d_Q, \tag{58}$$

where d_Q captures flow-dependent disturbances from valve switching. A PI controller is

$$u_p = K_c e + K_i \int e\,\mathrm{d}t, \qquad e = p_m^\star - p_m. \tag{59}$$

For the nominal disturbance-free plant, the characteristic polynomial is

$$\tau_p s^2 + (1 + K_p K_c)s + K_p K_i = 0. \tag{60}$$

Theorem 1 (Nominal PI stability)**.** *For $\tau_p > 0$, $K_p > 0$, $K_c > -1/K_p$, and $K_i > 0$, the nominal closed-loop pressure subsystem is asymptotically stable.*

Proof. The second-order polynomial has positive coefficients under the stated conditions. By the second-order Routh-Hurwitz criterion, both roots have strictly negative real parts. □

11.2 Anti-windup and command scheduling

Because pump speed and valve positions saturate, the integral state must not continue accumulating during saturation. A standard back-calculation law is

$$\dot{z} = e + K_{aw}(u_{sat} - u_{raw}), \tag{61}$$

with $u_{raw} = K_c e + K_i z$. Feedforward can improve response when multiple heads are activated:

$$u_{ff} = \hat{f}^{-1}(p_m^\star, \sum_i s_i A_i), \tag{62}$$

where $\hat{f}^{-1}$ is an experimentally fitted inverse pump/manifold map.

12 Supervisory Cycle Control

12.1 State machine

A robust wash cycle is naturally expressed as a guarded state machine rather than a fixed timer. One possible sequence is:

$$\texttt{Idle} \rightarrow \texttt{Position} \rightarrow \texttt{Precheck} \rightarrow \texttt{Wet} \rightarrow \texttt{Dose} \rightarrow \texttt{Massage} \rightarrow \texttt{Rinse} \rightarrow \texttt{Drain} \rightarrow \texttt{Finish}. \tag{63}$$

Every transition requires explicit guards. For example, `Precheck`→`Wet` requires valid temperature, adequate reservoir level, open drain path, acceptable head pose, and no active fault.

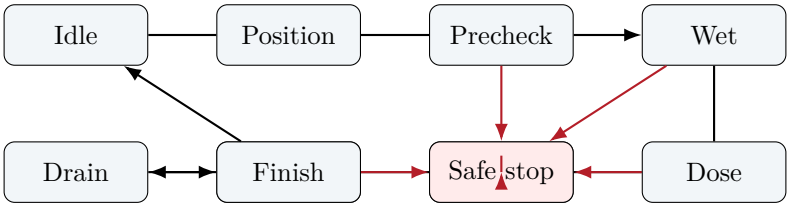


Figure 11: Guarded treatment state machine. Safety transitions have higher priority than nominal phase transitions.

12.2 Wash-phase logic

A state should terminate when its physical objective is achieved, not only when a timer expires. Example conditions are

$$\texttt{Wet}: \quad \min_{\boldsymbol{x}\in\mathcal{S}} C(\boldsymbol{x}) \geq C_{wet}^{min}, \tag{64}$$

$$\texttt{Dose}: \quad D_s = \int c_s(t)Q(t)\mathrm{d}t \geq D_s^{target}, \tag{65}$$

$$\texttt{Massage}: \quad A_m = \int \phi(F_n, v_t)\mathrm{d}t \geq A_m^{target}, \tag{66}$$

$$\texttt{Rinse}: \quad \hat{c}_m \leq c_{rinse}^{max} \text{ and } C_{rinse} \geq C_r^{min}. \tag{67}$$

13 Model Predictive Cycle Optimization

The main opportunity for L Pro is to use its distributed heads as a controllable field rather than drive all actuators identically. At sampling time k, an MPC problem over horizon H can minimize

$$J = \sum_{j=0}^{H-1} \Big[w_c \left\| \boldsymbol{C}_{k+j} - \mathbf{1} \right\|_W^2 + w_p (p_{k+j} - p^\star)^2 + w_T (T_{k+j} - T^\star)^2 \tag{68}$$

$$+ w_u \left\| \boldsymbol{u}_{k+j} \right\|_2^2 + w_{\Delta u} \left\| \boldsymbol{u}_{k+j} - \boldsymbol{u}_{k+j-1} \right\|_2^2 \Big] + w_f \left\| \boldsymbol{C}_{k+H} - \mathbf{1} \right\|_W^2 . \tag{69}$$

The constraints include

$$p_{min} \le p_m \le p_{max}, \tag{70}$$
$$T_{min} \le T \le T_{max}, \tag{71}$$
$$0 \le u_i \le 1, \tag{72}$$
$$F_{n,i} \le F_{max,i}, \tag{73}$$
$$h_b \le h_{crit}, \tag{74}$$
$$Q_d \ge Q_{in} - \epsilon_h. \tag{75}$$

Only the first control action is applied, after which the estimator updates the state and the optimization repeats.

13.1 Water-energy-time Pareto objective

Total water consumption is

$$V_w = \int_0^T Q_w(t)\mathrm{d}t, \tag{76}$$

while hydraulic pump energy is approximately

$$E_p = \int_0^T \frac{\Delta p(t) Q(t)}{\eta_p(t)}\mathrm{d}t, \tag{77}$$

and heater energy is

$$E_h = \int_0^T P_h(t)\mathrm{d}t. \tag{78}$$

A scalar cycle objective is then

$$J_{cycle} = w_T T + w_V V_w + w_E(E_p + E_h) + w_Q \sum_m (1 - C_m(T))^2. \tag{79}$$

The weights can be product-tuned for fast, eco, or intensive modes while maintaining identical hard safety constraints.

14 Safety Constraints as Control Invariants

14.1 Safety hierarchy

The supervisory design should distinguish (i) hard invariants that must never be violated, (ii) soft comfort constraints, and (iii) performance objectives. A representative hierarchy is:

1. **Hard:** no unsafe temperature, no pressure above certified bound, no unbounded basin level, no motion into protected geometry, no liquid delivery when required sensing is invalid.
2. **Comfort:** limited pressure slew, bounded massage force, gradual temperature transitions, limited acoustic/vibration transients.
3. **Performance:** coverage, cleaning surrogate, cycle time, water use, energy.

Hard constraints should be enforced below the optimization layer by independent monitors whenever possible.

14.2 Barrier-function formulation

For a safety quantity $h(\boldsymbol{x}) \geq 0$, a control barrier condition is

$$\dot{h}(\boldsymbol{x}, \boldsymbol{u}) + \alpha(h(\boldsymbol{x})) \geq 0, \tag{80}$$

where α is an extended class-$\mathcal{K}$ function. For an upper pressure limit,

$$h_p = p_{max} - p_m. \tag{81}$$

The controller can project the nominal command $\boldsymbol{u}_{nom}$ onto the safe set:

$$\boldsymbol{u}_{safe} = \arg\min_{\boldsymbol{u}} \ \|\boldsymbol{u} - \boldsymbol{u}_{nom}\|_2^2 \tag{82}$$

$$\text{s.t. } \dot{h}_j(\boldsymbol{x}, \boldsymbol{u}) + \alpha_j h_j(\boldsymbol{x}) \geq 0, \quad \forall j. \tag{83}$$

This separates performance optimization from last-line constraint enforcement.

14.3 Two-channel emergency response

A fault that could increase delivered energy should cause an energy-removing action: close inlet valves, disable heater, stop massage motion, and continue drainage if drainage itself is healthy. This asymmetry is important: safe stop need not mean "turn everything off" if an active drain is required to remove residual water.

15 Fault Detection and Graceful Degradation

15.1 Residual-based detection

For residual vector $\boldsymbol{r}_k = \boldsymbol{y}_k - \hat{\boldsymbol{y}}_k$ with covariance S_k, the normalized innovation statistic is

$$\Gamma_k = \boldsymbol{r}_k^T S_k^{-1} \boldsymbol{r}_k. \tag{84}$$

Under nominal Gaussian assumptions, Γ_k approximately follows a χ^2 distribution. Persistent threshold violations can trigger fault isolation.

15.2 Example fault signatures

Table 2: Physics-based fault signatures.

Fault	**Expected signature**	**Safe response**
Blocked nozzle	Manifold pressure high, branch flow low	Disable branch; redistribute coverage
Leak / detached tube	Pump command high, pressure low, unexplained tank loss	Close source valve; drain; alarm
Drain blockage	Basin level rising despite drain command	Stop inlet; maximize drain; request service
Temperature sensor fault	Inconsistent redundant temperatures or implausible thermal residual	Heater off; use cold-safe rinse or terminate
Contact overload	Force above limit or rapid force derivative	Retract/relax treatment head
User motion	Head-pose residual / contact pattern changes abruptly	Pause jets and motion; re-register geometry
Dosing fault	Metering flow inconsistent with concentration model	Stop additive; purge/rinse

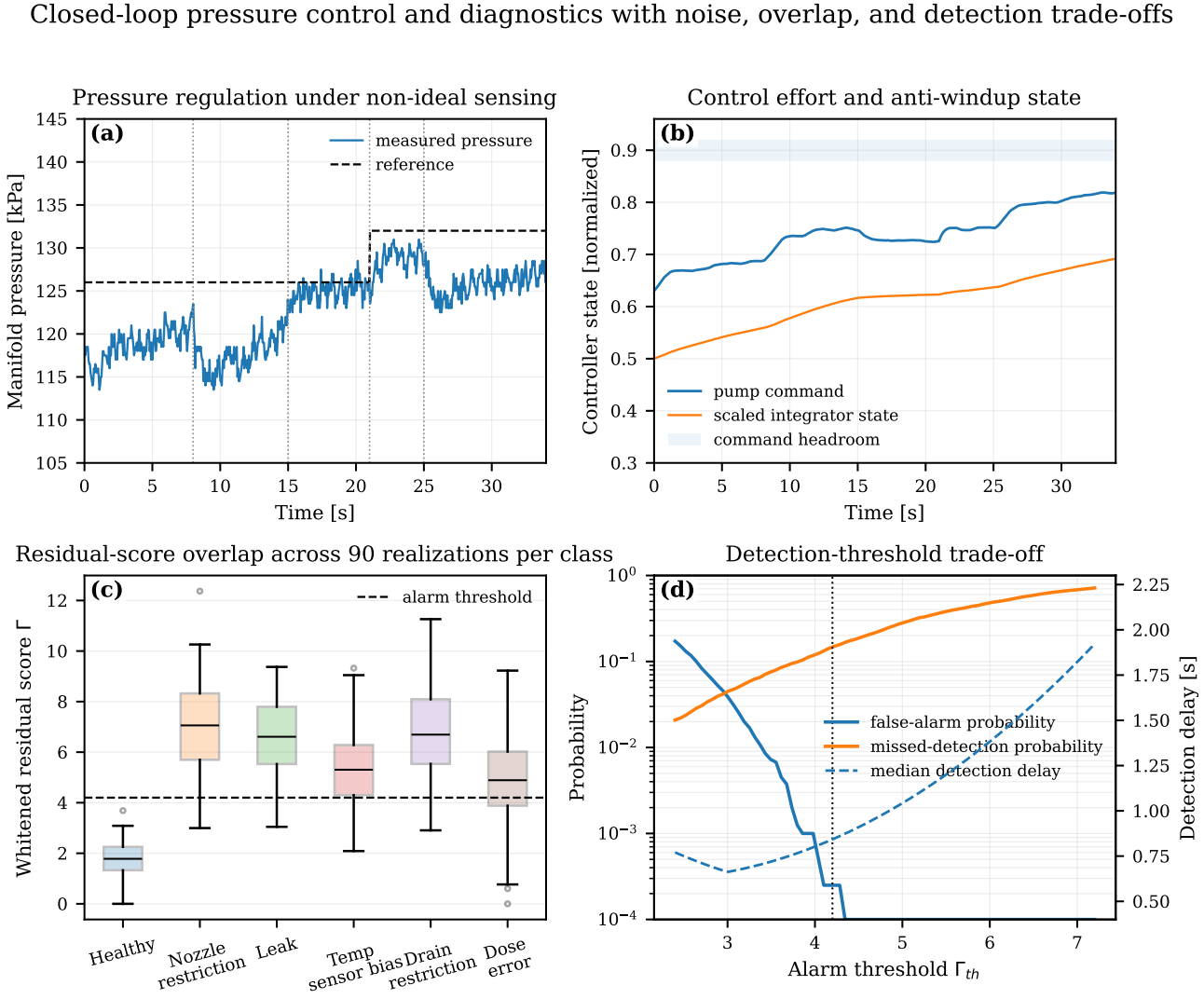


Figure 12: Integrated pressure-control and diagnostics study with non-ideal sensing and class overlap. (a) Pressure regulation includes multiple hydraulic switching events, pump ripple, sensor noise, and a set-point change. (b) Pump effort and the anti-windup integrator show the control cost of those disturbances. (c) Residual-score distributions for healthy and faulted cases overlap, as expected in a realistic detector. (d) The threshold sweep exposes the false-alarm, missed-detection, and detection-delay trade-off instead of assuming perfectly separable fault signatures.

The distributed geometry offers useful graceful degradation. If one wash head is disabled, the planner can increase exposure from neighboring heads while maintaining local pressure constraints. This redundancy can be quantified with a coverage matrix $A \in \mathbb{R}^{M \times N}$. Removing head j deletes column j. A useful robustness metric is

$$\kappa_{cov} = \min_j \frac{\max_{\boldsymbol{u} \in \mathcal{U}_{-j}} \min_m (A_{-j}\boldsymbol{u})_m}{\max_{\boldsymbol{u} \in \mathcal{U}} \min_m (A\boldsymbol{u})_m}. \tag{85}$$

Values near one imply strong single-head fault tolerance.

16 Human Positioning and Relative Motion

16.1 Head pose as the controlled geometry

The relevant geometric variable is not machine pose alone but the relative transform from machine frame $\{M\}$ to head frame $\{H\}$:

$$ {}^{M}T_{H} = \begin{bmatrix} {}^{M}R_{H} & {}^{M}\boldsymbol{p}_{H} \\ 0 & 1 \end{bmatrix}. \tag{86} $$

A nozzle point ${}^{M}\boldsymbol{o}_{i}$ transforms into head coordinates by

$$ {}^{H}\boldsymbol{o}_{i} = ({}^{M}R_{H})^{T}({}^{M}\boldsymbol{o}_{i} - {}^{M}\boldsymbol{p}_{H}). \tag{87} $$

Even if the treatment heads are mostly fixed, measuring this transform allows the controller to compute standoff and incidence accurately.

16.2 Positioning arms

The side articulated modules visible in the frontal and side renders can be treated as user positioning or accessory-support mechanisms. For a serial arm with generalized coordinates $\boldsymbol{q}$, end-effector velocity is

$$ \dot{\boldsymbol{x}} = J(\boldsymbol{q})\dot{\boldsymbol{q}}. \tag{88} $$

A damped least-squares command avoids excessive joint velocity near singularities:

$$ \dot{\boldsymbol{q}} = J^{T}(JJ^{T} + \lambda^{2}I)^{-1}\dot{\boldsymbol{x}}^{\star}. \tag{89} $$

If these modules are passive rather than motorized in production, the same geometry remains useful for ergonomic analysis rather than closed-loop actuation.



(a) Positioning modules deployed. (b) Side view of head interface and arm projection.

Figure 13: External geometry relevant to user positioning and protected-zone clearance.

17 Cleaning-Effectiveness Surrogate Model

Directly measuring "cleanliness" online is difficult, so the controller needs a surrogate based on observable exposure. Define normalized hydraulic, chemical, and mechanical doses at location m:

$$\bar{H}_m = \frac{H_m}{H_{ref}}, \tag{90}$$

$$\bar{D}_m = \frac{1}{D_{ref}} \int_0^T c_s q_m \mathrm{d}t, \tag{91}$$

$$\bar{M}_m = \frac{1}{M_{ref}} \int_0^T \phi(F_{n,m}, v_{t,m}) \mathrm{d}t. \tag{92}$$

A saturating quality surrogate is

$$S_m = 1 - \exp\left[-(\alpha_H \bar{H}_m + \alpha_D \bar{D}_m + \alpha_M \bar{M}_m + \alpha_{HD} \bar{H}_m \bar{D}_m)\right]. \tag{93}$$

The interaction term captures the fact that detergent is more effective when sufficiently transported through wet hair. The overall objective can use a risk-sensitive aggregation

$$S_{global} = \frac{1}{M}\sum_{m=1}^{M} S_m - \lambda_S \,\mathrm{Var}(S_m), \tag{94}$$

so spatial nonuniformity is penalized rather than hidden by a high mean.

The coefficients in Eq. (93) should not be guessed for production. They are parameters to identify from controlled wash tests with repeatable soil loading or optical/chemical residual measurements.

18 Stochastic Digital-Twin Design Study

The numerical study is intentionally *not* constructed as a set of smooth, best-case response curves. Instead, the analytical models above are embedded in a stochastic scenario generator. For scenario s, uncertain physical and sensing parameters are sampled as

$$\boldsymbol{\theta}^{(s)} \sim p(\boldsymbol{\theta}), \qquad \boldsymbol{y}^{(s)}(t) = \mathcal{F}\Big(\boldsymbol{u}(t), \boldsymbol{\theta}^{(s)}, \boldsymbol{d}^{(s)}(t)\Big) + \boldsymbol{\epsilon}^{(s)}(t), \tag{95}$$

where $\mathcal{F}$ is the coupled hydraulic–thermal–contact model, $\boldsymbol{d}$ contains disturbances such as hair loading, valve switching, and user alignment, and $\boldsymbol{\epsilon}$ represents sensor noise and quantization. Reported bands and boxplots are therefore distributions over simulated operating conditions rather than visually clean nominal trajectories.

The computational perturbations include pump-curve gain and hydraulic resistance, effective nozzle area, branch-to-branch variation, head alignment, local hair resistance and shielding, contact stiffness, metering gain, thermal inlet temperature, actuator delay, and sensor noise. Their magnitudes are chosen to create a demanding but physically interpretable design study; they must be replaced by tolerance-stack, calibration, and prototype distributions before any external performance claim is made.

A robust controller is evaluated using both expected performance and a chance constraint

$$\Pr[g_j(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\theta}) \le 0] \ge 1 - \alpha_j, \tag{96}$$

so an operating point is not considered acceptable merely because its mean trajectory is safe. The figures use repeated scenarios, lower-tail spatial metrics, false-alarm/missed-detection trade-offs, and constraint-violation probability for precisely this reason.

18.1 Pressure-flow operating envelope

A more realistic pump/system pair is written

$$\Delta p_p = H_0 u_p^2 - K_p Q^2, \tag{97}$$

$$\Delta p_s = p_0 + K_s Q^2, \tag{98}$$

where H_0, K_p, and K_s are uncertain rather than fixed. The equilibrium is

$$Q^\star = \sqrt{\frac{H_0 u_p^2 - p_0}{K_p + K_s}}, \tag{99}$$

when the numerator is positive. Branch-flow safety is then evaluated through Eq. (96), e.g., $\Pr(Q_i^{min} \leq Q_i \leq Q_i^{max}) \geq 0.95$. This produces the operating clouds and probability contours in Fig. 5 rather than a single exact intersection.

The hydraulic power still scales as $P_h = \Delta p\, Q/\eta_p$, so raising pressure to compensate for every adverse load realization is energetically expensive and can consume actuator headroom. That coupling motivates flow balancing and coverage-aware scheduling instead of globally increasing pump command.

18.2 Coverage scheduling benefit

Consider $M = 100$ scalp cells and a planner that always activates the least-covered reachable region. If mean normalized exposure rate is constant, uniform scheduling reduces the variance penalty in S_{global} without increasing total water. The simple analytical bound from Eq. (17) is

$$\min_m C_m(T) \geq 1 - \exp(-H_{min}(T)). \tag{100}$$

Thus guaranteeing $H_{min} \geq 3$ guarantees $C_m \geq 0.9502$ for every modeled cell. Rather than judging a controller by a single curve, the more relevant engineering question is how complete candidate cycles populate the multi-objective design space in water, time, energy, spatial coverage, and robustness.

Table 3: Scenario populations used in the stochastic design study. Counts describe computational realizations, not experimental subjects or prototype repetitions.

Study block	**Scenarios**	**Dominant randomized effects**	**Reported statistic**
Hydraulics	180 + 80/case	Pump gain, hydraulic resistance, branch tolerance, sensing ripple	Operating cloud, branch-flow boxplots, chance constraint
Coverage	120 + 150	Head alignment, hair resistance, local shielding, spray footprint	Lower-tail exposure, spatial CV, 10–90% envelope
Contact	60/module	Effective stiffness, module gain, cyclic ripple, transient strand interaction	Force dispersion, hysteresis, slip-risk map
Dosing	18–50	Dead-volume fraction, metering gain, pulsation, sensor quantization	Flush tail, calibration band, cycle repeatability
Thermal	90	Inlet temperature, time constant, transport delay, flow disturbances	Scenario envelope, saturation, tail metrics
Diagnostics	90/class	Residual noise, fault magnitude, sensor overlap	Score distributions, false alarm / miss / delay trade-off
Cycle design	520	Time, flow command, adaptive scheduling strength, constraint risk	Robust Pareto set and policy uncertainty

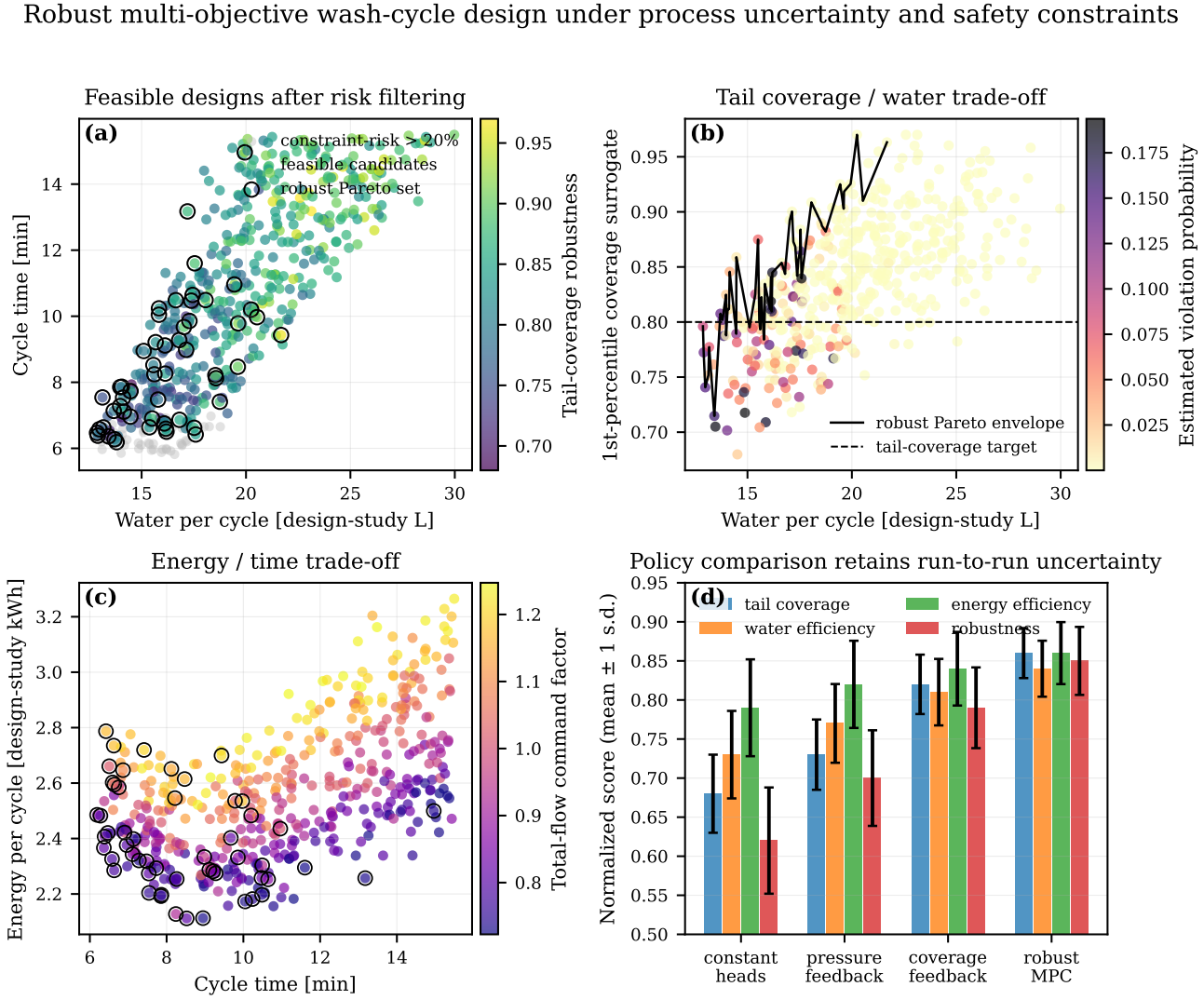


Figure 14: Robust multi-objective cycle-design study using 520 randomized candidate schedules. (a) Candidates with estimated safety-constraint risk above 20% are separated from the feasible design cloud, and the robust Pareto set is outlined. (b) Tail coverage is plotted against water use and colored by estimated violation probability. (c) The energy–time surface retains the effect of total-flow command. (d) Policy-level bars report mean ± one standard deviation rather than deterministic scores. The dimensional values are synthetic design-study outputs used to exercise the optimization structure, not experimental L Pro performance claims.

18.3 Rinse concentration decay

From Eq. (38), when additive input is shut off and clean water enters, concentration decays as

$$c_m(t) = c_m(0)e^{-t/\tau_m}. \tag{101}$$

Five mixer time constants reduce residual concentration to $e^{-5} \approx 0.0067$, giving an intuitive basis for rinse timing.

19 Mechanical Packaging and Serviceability

The L Pro renders show a notable separation between user-facing surfaces and service-facing internals. Large removable exterior panels conceal pumps, hoses, filters, reservoir mounts, and electronic modules. This architecture can reduce routine service time if the internal fluid path is designed around modules rather than point-to-point plumbing.



(a) Upper exterior. (b) Lower/base exterior. (c) Side panelization. (d) Rear interface.

Figure 15: Exterior panel strategy. Service seams and modular lower-body panels are visible while user-facing surfaces remain smooth and washable.

A serviceability objective can be formalized. Let component j fail with rate λ_j, require service time t_j, and require removal of n_j access modules. Expected service burden is

$$J_{service} = \sum_j \lambda_j \left(w_t t_j + w_n n_j + w_f t_{flush,j} \right). \tag{102}$$

This metric encourages high-failure-rate consumables such as filters, cartridges, and wetted seals to be reachable without disturbing major structural assemblies.

20 Sanitation and Cross-Contamination Modeling

A personal-care fluid system must be designed so that residual liquid does not persist indefinitely in dead legs. Consider a fluid volume V_d that is flushed by clean water at flow Q_f. Under well-mixed assumptions, contaminant concentration satisfies

$$\dot{c}_d = -\frac{Q_f}{V_d} c_d, \tag{103}$$

so

$$c_d(t) = c_d(0) e^{-Q_f t / V_d}. \tag{104}$$

To reach a specified residual ratio $r = c_d/c_d(0)$, the required flush volume is

$$V_{flush} = Q_f t = -V_d \ln r. \tag{105}$$

For example, a target $r = 0.01$ requires $4.605V_d$ under the ideal-mixing model. Plug-flow segments require less volume; stagnant pockets require more. This makes dead-volume minimization a direct water-efficiency design objective.

21 Acoustic and Vibration Considerations

User comfort also depends on pump, valve, and actuator noise. If multiple periodic sources have RMS pressure amplitudes p_i, incoherent acoustic levels combine approximately through energy:

$$p_{rms,total} = \sqrt{\sum_i p_i^2}. \tag{106}$$

For structural vibration, a local single-degree-of-freedom mount can be modeled as

$$m\mathrm{d}x + c\dot{x} + kx = F_0 \sin \omega t. \tag{107}$$

The displacement transmissibility magnitude is

$$|H(j\omega)| = \frac{1}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}. \tag{108}$$

Pump mounts should place the dominant operating frequency away from lightly damped enclosure modes. Variable-speed control can additionally avoid persistent operation at a measured resonance.

22 Design Rules Derived from the Model

The multiphysics model leads to several concrete engineering rules:

1. **Calibrate each branch.** Because nozzle area errors enter flow linearly while pressure enters as a square root, per-head gain correction can materially improve uniformity.
2. **Plan exposure, not only actuator time.** The saturation law $C = 1 - e^{-H}$ makes repeated treatment of already-covered areas inefficient.
3. **Keep drainage overprovisioned.** The drain path should have a certified margin above worst-case simultaneous inlet flow, not merely equal nominal flow.
4. **Use compliance for geometry uncertainty.** Compliant heads convert small pose errors into bounded force changes instead of rigid collision loads.
5. **Control chemical ratio.** Dosing should scale with instantaneous water flow so concentration remains stable when hydraulic commands change.
6. **Treat temperature as a dynamical state.** Heater power alone is not a reliable proxy for outlet temperature during changing flow.
7. **Separate hard safety from optimization.** A low-level invariant monitor should be able to override MPC or phase logic.
8. **Exploit spatial redundancy.** The distributed head array allows reallocation around a failed or uncomfortable head.

9. **Minimize wetted dead volume.** Every added dead leg increases sanitation flush volume according to Eq. (105).

23 Calibration and Experimental Validation Plan

A publishable experimental version of this work should replace illustrative parameters with measured L Pro data. The minimum calibration program is summarized in Table 4.

Table 4: Recommended L Pro model-calibration program.

Experiment	Input sweep	Measured outputs	Identified parameters
Pump/manifold bench	Pump speed, valve count, tank level	Manifold pressure, branch flows, electrical power	Pump curve, K_h, valve gains
Nozzle jet bench	Pressure, standoff, angle	Flow, jet diameter, force plate pressure	C_d, k_e, C_i
Contact bench	Indentation depth and speed	Normal force, hysteresis	E^*, k_c, b_c
Thermal bench	Flow, inlet temperature, heater power	Outlet temperature transient	UA, m, τ_T
Mixer bench	Water/additive step commands	Concentration or conductivity proxy	V_m, transport delay
Drain bench	Inlet flow and hair/filter loading	Basin level, drain rate	$Q_d(h)$, clog model
Coverage phantom	Head geometry, nozzle sequence	Surface wetting map	σ_i, χ_i, coverage model
Integrated wash test	Cycle recipe and hair class	Water, energy, time, residual soil, comfort scores	Quality surrogate coefficients
Fault injection	Blockage, leak, sensor bias, user motion	Residuals, detection delay	Fault thresholds, safe-stop timing

23.1 Coverage phantom

A particularly valuable test fixture is a scalp-shaped phantom instrumented with absorbent or moisture-sensitive patches. If measured wetness at cell m is w_m, fit the model parameters by

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{m=1}^{M} \left[w_m - \hat{w}_m(\boldsymbol{\theta}) \right]^2 + \lambda \left\| \boldsymbol{\theta} - \boldsymbol{\theta}_0 \right\|^2 . \tag{109}$$

The same fixture can evaluate single-head failure and head-pose perturbations reproducibly.

23.2 Statistical reporting

For each calibrated performance quantity y, report mean, standard deviation, and confidence interval across repeated trials. For example,

$$\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i, \qquad s_y = \sqrt{\frac{1}{n-1} \sum_i (y_i - \bar{y})^2}, \tag{110}$$

and for approximately normal samples a two-sided confidence interval is

$$\bar{y} \pm t_{1-\alpha/2,n-1} \frac{s_y}{\sqrt{n}}. \tag{111}$$

No illustrative curve in this document should be converted into a product performance claim until such measurements are completed.

24 Implementation Blueprint

A practical embedded implementation can use a fast inner loop and slower outer loops:

- **0.5–2 kHz:** motor commutation, force or local actuator control if required.
- **50–200 Hz:** pressure, flow, contact-force, and basin-level safety control.
- **5–20 Hz:** EKF state estimation, head-pose update, branch balancing.
- **1–5 Hz:** coverage map update, MPC replanning, phase objectives.
- **Event driven:** UI, recipe selection, logging, maintenance diagnostics.

These ranges are architectural examples, not final production requirements.

Listing 1: High-level treatment-cycle pseudocode.

```
initialize_models_and_calibration()
while user_session_active:
    read_sensors()
    estimate_state()
    update_fault_residuals()

    if hard_safety_violation():
        close_inlet_valves()
        disable_heater_and_contact_motion()
        run_drain_if_healthy()
        enter_safe_stop()
        continue

    update_head_geometry_and_coverage_map()
    phase = supervisory_state_machine()

    if phase in {WET, DOSE, MASSAGE, RINSE}:
        target = formulate_phase_objective(phase)
        u_nom = solve_constrained_mpc(target)
        u_safe = safety_filter(u_nom)
        apply_commands(u_safe)
    else:
        apply_phase_specific_commands(phase)

    log_state_and_diagnostics()
```

25 Limitations

The current document is intentionally a theory-first paper. The supplied renders reveal packaging and component classes but not exact component ratings or measured behavior. Consequently, four limitations must be stated explicitly. First, treatment-head internal geometry is not known; the jet and contact models therefore use general orifice, free-jet, and compliant-contact laws. Second, no prototype calibration data were supplied; numerical plots are illustrative. Third, the biological interaction between hair, scalp, surfactant, and massage is simplified into a surrogate exposure model. Fourth, formal product safety certification requires implementation-specific risk analysis, electrical/liquid ingress verification, materials testing, and human-subject validation beyond the mathematical invariants developed here.

These limitations do not reduce the value of the framework. They define exactly which measurements are needed to convert the document from a design-theory paper into a validated system paper.

26 Conclusion

L Pro can be understood as a closed-loop distributed treatment system whose core challenge is to deliver spatially uniform hydraulic, chemical, thermal, and mechanical exposure while maintaining strict human-interface constraints. The visible architecture supports this interpretation: a dense array of treatment heads acts on the scalp; pumps, reservoirs, cartridges, manifolds, and drains establish the fluid loop; compliant interfaces manage geometric uncertainty; and the enclosure packages the machine into a cleanable, serviceable appliance.

The derived equations provide a rigorous operational explanation. Pump and nozzle equations connect actuator command to flow; jet models connect flow to local momentum; exposure kernels convert distributed actuation into coverage; contact mechanics bound massage pressure; mixing and thermal balances govern chemistry and comfort; drainage inequalities prevent accumulation; estimation and fault residuals detect abnormal behavior; and constrained MPC coordinates the complete cycle. Several results provide useful design guarantees before full hardware validation: coverage is monotone under nonnegative exposure, a sufficient drain margin prevents basin accumulation, the thermal interval can be made forward invariant, and the nominal pressure-control loop is stable under simple gain conditions.

The next engineering step is therefore not to add more equations, but to identify the model coefficients on the actual L Pro prototype. With branch-flow, jet-force, thermal, contact, mixing, and coverage calibration, the same framework becomes a predictive digital twin that can optimize recipes, diagnose faults, compare hardware revisions, and quantify water-energy-time tradeoffs with defensible data.

A Detailed Derivations

A.1 Combined pump-system operating point

Suppose the pump curve at normalized command u is

$$\Delta p_p = a_0 u^2 - a_1 Q^2 \tag{112}$$

and the downstream system is

$$\Delta p_s = p_0 + b_1 Q + b_2 Q^2. \tag{113}$$

Equating them gives

$$(a_1 + b_2)Q^2 + b_1 Q + (p_0 - a_0 u^2) = 0. \tag{114}$$

The physically meaningful positive root is

$$Q(u) = \frac{-b_1 + \sqrt{b_1^2 - 4(a_1 + b_2)(p_0 - a_0 u^2)}}{2(a_1 + b_2)}. \tag{115}$$

Feasibility requires a nonnegative discriminant. This expression can seed pump feedforward or expose low-tank/high-resistance conditions where a desired operating point is physically unreachable.

A.2 Pressure-to-impact sensitivity

Combining Eqs. (7), (11) and (12), and using $v_0 = Q/A_n$, yields approximately

$$p_{imp} \propto \frac{\Delta p}{(1 + k_e z/D_n)^2} \cos^2\theta. \tag{116}$$

Thus the logarithmic sensitivity is

$$\frac{\mathrm{d}p_{imp}}{p_{imp}} \approx \frac{\mathrm{d}\Delta p}{\Delta p} - \frac{2k_e\,\mathrm{d}z/D_n}{1 + k_e z/D_n} - 2\tan\theta\,\mathrm{d}\theta. \tag{117}$$

This gives a local error budget for pressure, head-position, and angular uncertainty.

A.3 Minimum exposure time

If a point receives constant aggregate exposure rate λ, then $H = \lambda T$ and

$$C(T) = 1 - e^{-\lambda T}. \tag{118}$$

To guarantee target $C^\star$,

$$T \geq -\frac{1}{\lambda}\ln(1 - C^\star). \tag{119}$$

Doubling exposure rate halves the required time only in this ideal unsaturated-input model; in the real machine the allowed rate is limited by pressure, drainage, and comfort.

A.4 Discrete thermal model

For sample period Δt, write Eq. (42) as

$$\dot{T} = -aT + b, \quad a = \frac{\dot{m}c_p + UA}{mc_p}, \quad b = \frac{\dot{m}c_p T_{in} + P_h + UAT_a}{mc_p}. \tag{120}$$

Exact discretization gives

$$T_{k+1} = e^{-a\Delta t}T_k + \frac{1 - e^{-a\Delta t}}{a}b_k. \tag{121}$$

This form is preferable to forward Euler inside MPC when the sample period is not extremely small.

A.5 Tank level from volume

For a reservoir with cross-sectional area $A(h)$,

$$V(h) = \int_0^h A(\xi)\mathrm{d}\xi, \tag{122}$$

and therefore

$$\dot{h} = \frac{Q_{in} - Q_{out}}{A(h)}. \tag{123}$$

Nonuniform reservoir geometry should therefore use a calibrated $V(h)$ lookup rather than assume a linear level-to-volume map.

A.6 Water-hammer bound

Rapid valve closure can produce a transient pressure increase. A classical upper estimate is the Joukowsky relation

$$\Delta p_{wh} = \rho a \Delta v, \tag{124}$$

where a is acoustic wave speed in the compliant fluid-line system. Flexible hoses reduce effective a, but fast valves can still generate a significant spike. Command slew limits and accumulator compliance should therefore be considered even when steady-state pressure is safe.

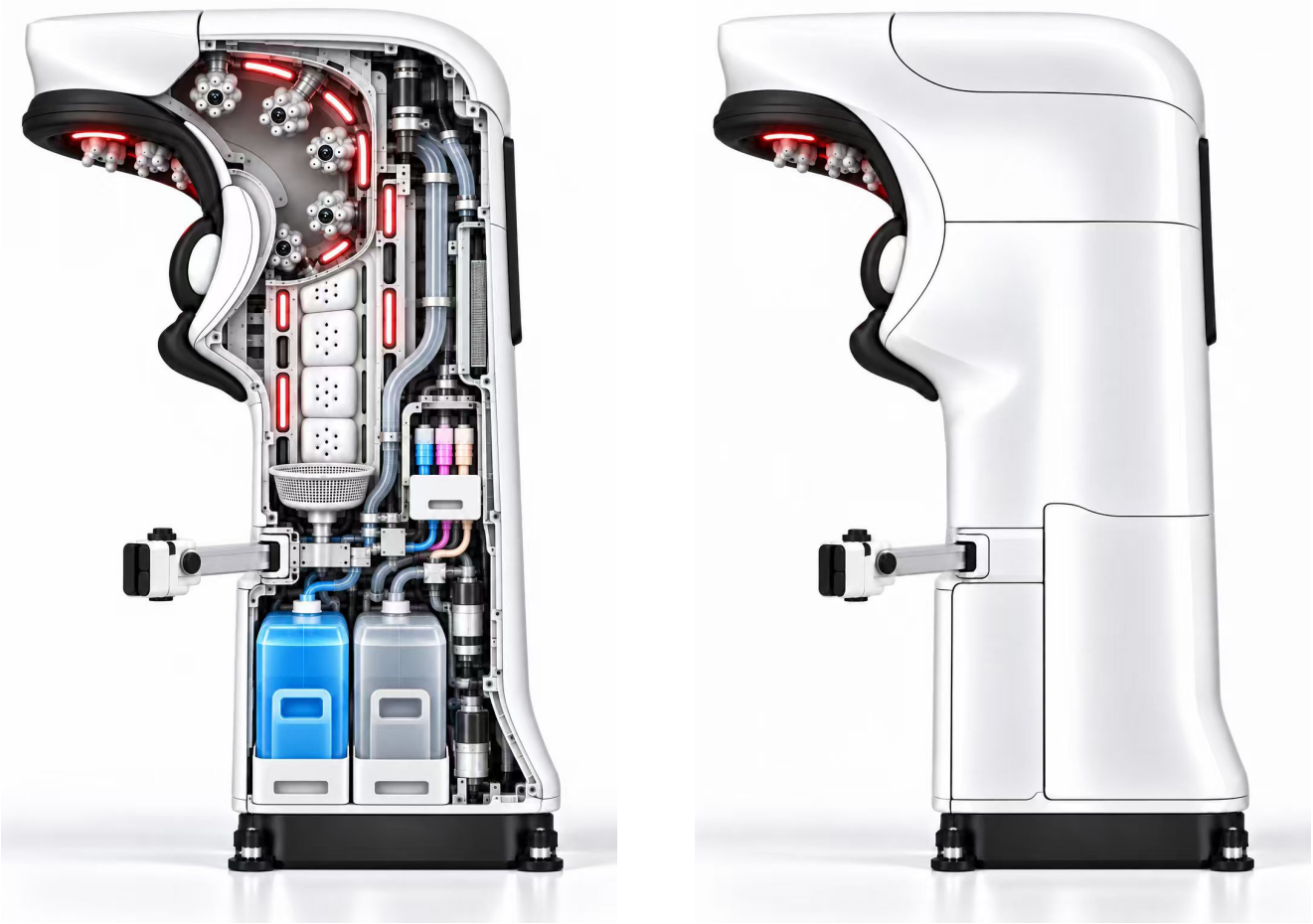
B Nominal Modeling Parameters for Simulation Only

The following table is deliberately dimensionless or symbolic where production values are unknown. It is intended to make the LaTeX source directly usable for simulation setup without implying certified L Pro specifications.

Table 5: Suggested model symbols and calibration sources.

Symbol	**Meaning**	**Unit**	**How to obtain**
C_d	Nozzle discharge coefficient	–	Flow/pressure bench fit
A_n	Effective nozzle area	m^2	CAD + flow fit
K_h	Aggregate hydraulic resistance	$Pa/(m^3/s)^2$	Pump/manifold sweep
k_e	Jet entrainment coefficient	–	Jet velocity/force map
C_i	Impact coefficient	–	Force plate
σ_i	Wetting kernel width	m	Coverage phantom
E^*	Effective contact modulus	Pa	Indentation test
k_c, b_c	Contact spring/-damper	N/m, Ns/m	Dynamic indentation
V_m	Mixing volume	m^3	Concentration step test
UA	Thermal loss coefficient	W/K	Thermal transient fit
$Q_d(h)$	Drain-flow map	m^3/s	Basin flow bench
R_k, Q_k	Sensor/process covariance	various	Repeated static/dynamic tests

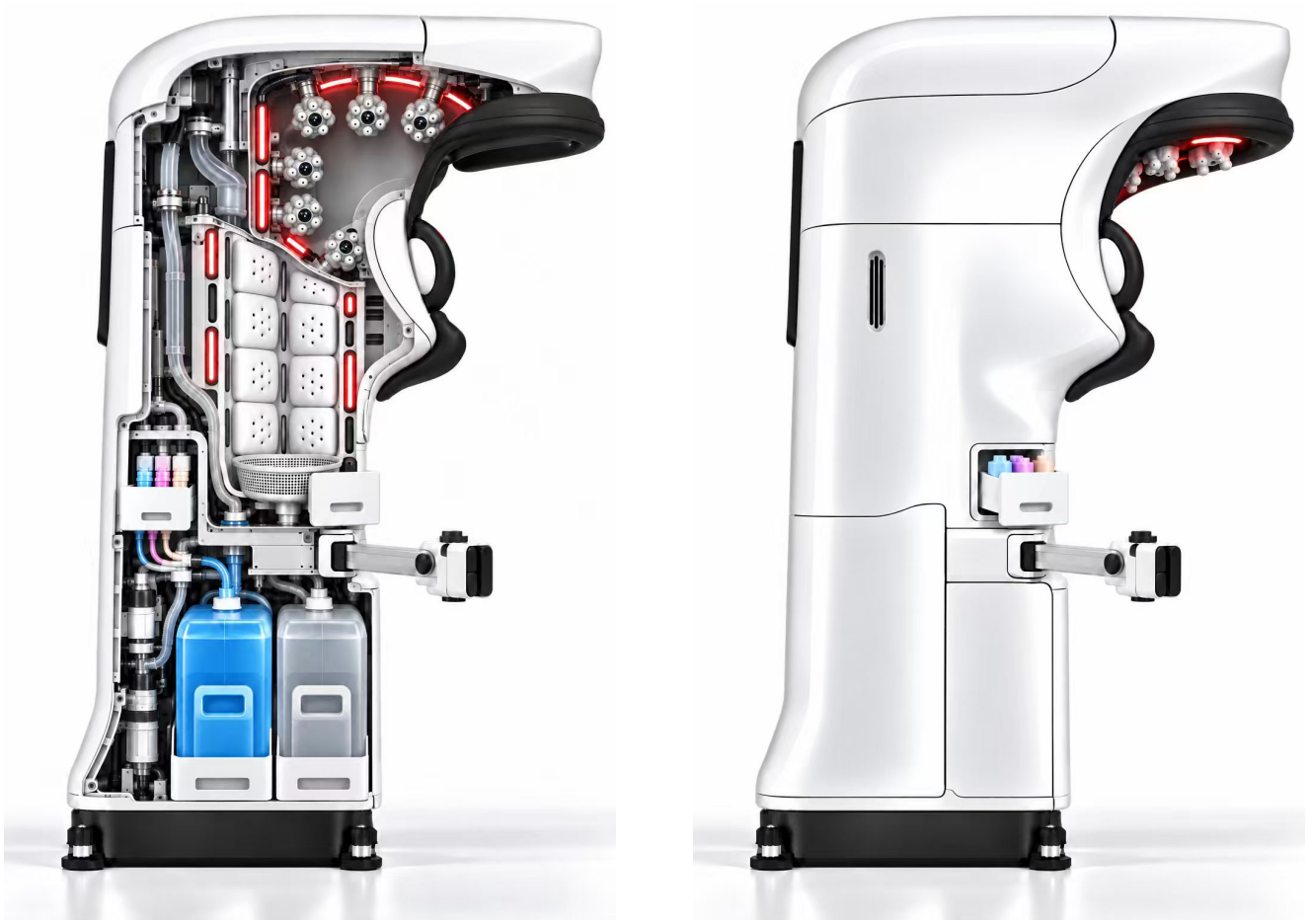
C Additional L Pro Render Plates



(a) Internal side A.

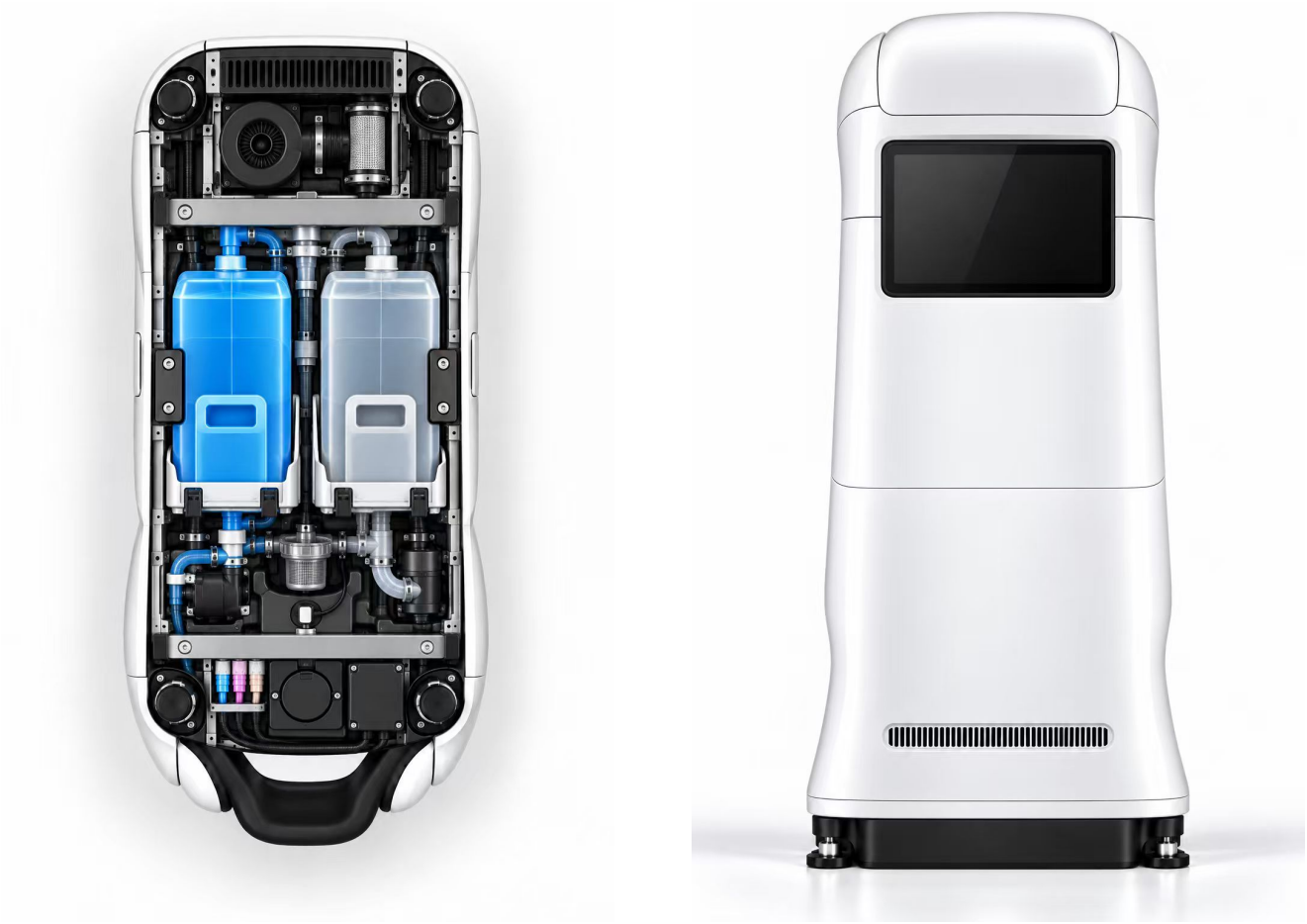
(b) External side A.

Figure 16: Matched side-A render pair.



(a) Internal side B. (b) External side B.

Figure 17: Matched side-B render pair.



(a) Internal rear/top view.

(b) External rear with display.

Figure 18: Rear packaging and user-interface relationship.

D Notation

Symbol	Definition
Q, Q_i	Total and branch volumetric flow rates
p_m	Main manifold pressure
ρ, μ	Liquid density and dynamic viscosity
C_d, A_n	Nozzle discharge coefficient and effective area
v_0, v_c	Nozzle exit and downstream jet centerline velocities
H	Accumulated local treatment exposure
C	Saturating coverage surrogate
F_n, F_t	Normal and tangential contact forces
E^*	Effective contact modulus
c_s, c_m	Delivered additive concentration and mixer concentration
T, T_{in}, T_a	Conditioned, inlet, and ambient temperatures
P_h	Heater power
V_b, h_b	Basin liquid volume and level
$\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{y}$	State, control, and measurement vectors

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